

The Impact of Women's Self Help Groups on Political Entry: Evidence from India

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Abstract

The democratization of political entry is essential for a robust representative democracy. We examine whether women's self-help groups (SHGs) – which expand financial access and social networks for disadvantaged groups – can spur political entry. Our study leverages the randomized rollout of an SHG program across 179 village councils (Gram Panchayats, GPs) in Bihar, India, where SHGs were introduced two years earlier in treatment GPs. The program targeted low-income, historically marginalized Scheduled Castes (SCs). Four years after SHGs are introduced, the number of SC candidates in treated GPs rises by 79% and SC women candidates by 127%. We find no effects on other types of candidates, including women or other marginalized castes. The effects on SC entry are concentrated in GPs reserved for SCs. New SC entrants are poorer and less educated. We find evidence for two mechanisms. First, SHGs provide leadership opportunities for group members. Second, SHG leaders mobilize voters, evidenced by a large increase in the number of votes polled for SC candidates. Consistent with a large candidate pool and the marginal candidate being of lower electability, the winner pool remains unchanged. Candidacy effects persist into subsequent election cycles, with some evidence of more SC women winning too, but we detect no changes in public-goods provision or workfare outcomes. Our study highlights the complementarity of both formal (reservations) and informal institutions (SHGs) in democratizing political entry.

JEL codes: D71, D72, O12

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I. Introduction

Leaders influence economic development and growth (Jones and Olken 2005). However, even when democratically elected, leaders tend to be overwhelmingly male and from elite backgrounds. For instance, in the United States, under 30% of 2023 senators were women, and nearly 80% of congress members were non-Hispanic white (Pew Research Center 2023). Thus, increasing representation from disadvantaged groups remains a global challenge for democracies. This is arguably more important for disadvantaged minorities in developing countries, who often face significant entry barriers. In India, the setting of this paper, women occupy 13% of seats in parliament (Ministry of Parliamentary Affairs, Government of India 2023).

Collective action is often seen as a means for under-represented groups to amplify their voice and influence decision-making (Banerjee, Iyer, and Somanathan 2007). One vehicle for collective action – women’s self-help groups (SHGs) – has gained popularity in the developing world (Brody et al. 2017; Gugerty, Biscaye, and Anderson 2019). SHGs are fora where poor rural women come together in small groups to collectively lend and borrow from each other. A vast literature shows that SHGs improve finance and livelihoods for women and their households, expand women’s social networks and empower them (Hoffmann et al. 2021; Kumar et al. 2019). Moreover, Prillaman (2023b) shows that SHGs also enhance women’s political participation in village meetings and other public platforms. Yet, the explicit connection between SHGs and political entry is unexamined.

In this paper, we study if SHGs spur political entry in local village-council – Gram Panchayat (“GP”) – elections in Bihar. Bihar, with over 120 million people, is one of India’s largest states and is also among its poorest. Its SHG program, Bihar Rural Livelihoods Project (more commonly known as “Jeevika”) was launched in 2006 and has established over 1 million SHGs to date, with over 12 million women.

While Jeevika was eventually rolled out to a majority of rural households, it prioritized low-income women, a majority of whom come from historically marginalized Scheduled Caste (SC) households (Sanyal, Rao, and Majumdar 2015).¹ SCs comprise a set of heterogeneous castes at the bottom of the caste hierarchy, facing severe discrimination and economic disadvantage (Thorat and Newman 2007; Pande 2003). These households experience poor health, education, and mortality outcomes (Deshpande 2011). As a result, political representation among SCs is extremely limited in open seats. Even in

1. Surendra (2024) documents the high take-up of Jeevika among SCs in Bihar and notes: “[J]eevika’s recruitment process ... targeted poor women, particularly those from Dalit [SC]/Adivasi [ST] households.” Here, we focus only on SCs and not Scheduled Tribes (STs) because there are very few STs in our sample. Our results are robust to including STs.

Bihar, which boasts a robust local democratic system – over 8,000 GPs, directly elected GP head positions, and quotas for women and SCs (called “reservations”) – nearly all SC winners come from reserved seats (Kumar and Sharan 2025).

We rely on the randomized controlled trial (RCT) evaluated in Hoffmann et al. (2021) to causally measure effects of exposure to SHGs on political entry of poor households and SCs in particular. The RCT, conducted by the World Bank alongside the Bihar government, randomized SHG presence over 179 GPs. SHGs were introduced two years earlier in treatment GPs in 2012 compared to control GPs in 2014. We measure effects on political entry in the 2016 GP elections.

We report several findings. First, treated GPs see an increase in SC candidates and SC women candidates by 79% and 127% respectively. To interpret the magnitude, we scale the experimental effect by the number of groups the program created, instrumenting groups formed with treatment assignment: one additional SC candidate emerges for every 22.2 groups, and one additional SC woman candidate for every 38.5. To put this in context, the average Bihar GP had just over 50 SHGs in 2015. However, we find no effects on non-SCs running in treatment GPs, including socially disadvantaged extremely backward castes (EBCs) or poor non-SC households. These findings suggest two things: first, that the program was successful at empowering SCs, who formed the vast majority of the core target group of poor households. Second, that there could exist institutional reasons for SCs to enter that may not be available to poor non-SCs.

Second, we examine whether the effects on SC political entry operate through a key institution designed to facilitate it: political reservations. We find that the effects on SCs are concentrated in SC-reserved GPs (i.e., *SHG + Reserved* GPs). In unreserved GPs, the estimated effect is small and statistically insignificant. By contrast, the interaction between SHG treatment and SC reservation yields a large and statistically significant effect for SC women: 3.7 additional candidates ($p = 0.088$), nearly ten times the mean of 0.38 SC women candidates in unreserved control GPs. Because reservation is not randomly assigned, we show that the result is robust to an empirical strategy that instruments for reservation with a threshold-crossing regression-discontinuity design, following Kumar and Sharan (2025).

Third, *SHG + Reserved* GPs encourage entry of candidates who are less electable *ex ante*. To test this, we match data on candidates with data on all citizens in experimental GPs using the 2011-12 Socioeconomic Caste Census (SECC), which allows us to obtain their household wealth and sub-caste size. Information on candidates’ education levels is obtained from their election affidavits. We combine the three into an electability index, weighting each by its ability to predict victory among control-group candidates. The

candidate pool in *SHG + Reserved* GPs scores about half a standard deviation lower on this index.

We provide evidence for two mechanisms. First, we show that new entrants are linked to SHG participation by looking at SHG-membership among candidates. Matching SHG rosters to candidate lists, we find that female candidates in treated GPs are twice as likely to be SHG members as those in control GPs. SHG leaders, none of whom contest in control GPs, also begin to run. These patterns do not merely reflect growth in the pool of SHG women: membership rises by half, but the member share among female candidates doubles, while leader candidacies increase roughly thirteen times as much as the number of leaders. The effect also extends to members' households. Matching candidates' households to the SECC, we find increased entry by both men and women.

Second, we look at turnout. Treatment does not increase turnout overall. However, two pieces of evidence suggest that SHGs allow SCs to mobilize more voters, especially in reserved GPs: first, votes polled rise by 9.3% in *SHG + Reserved* GPs relative to unreserved GPs; second, votes polled for SC candidates rise in these same GPs, while votes for SC candidates in unreserved GPs are unchanged. This suggests that SHG women may be leveraging their expanded networks to mobilize voters more effectively.

We next turn from candidacy to winning. We begin by looking at how treatment changes the composition of the winner class: across *all* GPs, we find no difference. Winners are no more – and no less – likely to be women, SCs, of lower *ex ante* electability or linked to SHGs in any way. Turning to *SHG + Reserved GPs*, where all the entry seems to be concentrated, we find no difference in winner quality either. Thus, SHGs help create a political class capable of entry, but given both the crowded pool – each candidate has an 8% chance of winning if randomly drawn – and the fact that marginal candidates are of lower electability, we see no effects on winning. Consistent with unchanged winners, we find no effects on public goods provision or NREGA participation.

While winner profiles remain unchanged in the short-run, we see some persistence in political entry. In the 2021 elections, more SC women continue to contest, though effect sizes are smaller. They are also more likely to win, especially in gender-open seats reserved for SCs, but there are only a handful of such seats. A plausible explanation for this persistence lies in program performance: treated GPs continue to have more SHGs than control GPs, and these groups remained better-functioning through 2024–25. We rationalize our findings using a simple framework that draws from Dal Bó and Finan (2018).

Our paper makes three broad contributions, to the literatures on political entry, political

reservations and SHGs.

We are the first paper to show an explicit link between SHGs and political entry. The current literature shows that SHGs have the potential to increase participation in local village government meetings, encourage women’s collective action, and consequently, through sustained political action, provide them with the means and agency to challenge existing patriarchal norms (Prillaman 2023a; Das, Maitra, and Sanyal 2019; Kumar et al. 2019). Our primary contribution is demonstrating that this increased agency (Sanyal, Rao, and Majumdar 2015) – especially in conjunction with reservations – extends to greater involvement in formal political participation in the form of contesting local elections. We deepen our contribution by delineating mechanisms: the democratization of the candidate pool caused by a lowering of entry barriers.

A vast literature studies how political reservations in local government fundamentally alter women’s and SC/STs political entry and consequent downstream outcomes (Chattopadhyay and Duflo 2004; Chin and Prakash 2011; Beaman et al. 2009; Kumar and Sharan 2025; Gulzar, Haas, and Pasquale 2023; Chauchard 2014). This paper highlights a complementary mechanism that amplifies the effects of formal institutional design²: informal collective organization through self-help groups (SHGs). We show SHG formation enhances the entry effects of SC reservation, jointly expanding the pool of entrants and reshaping who runs for office. In reserved GPs with SHGs, candidates tend to be less educated and from poorer households, suggesting that informal mobilization can deepen the inclusiveness of formal affirmative action. In doing so, the paper contributes to a growing literature on the democratic entry of different types of households into electoral politics under reservation (Bamezai et al. 2024).

Our research also contributes to a broader literature studying SHGs and their impacts on participating individuals and the community (Díaz-Martín et al. 2022). SHGs have been shown to improve a wide range of financial outcomes, including access to credit with lower interest rates, boosting savings, and boosting household income through various channels, including wage labor, self-employment, participation in NREGA, and providing a cushion against weather shocks (Deshpande, Khanna, and Walia 2023; Deshpande and Khanna 2021; Demont 2022; Hoffmann et al. 2021; Kumar et al. 2019; Pandey and Gupta 2019). Importantly, the literature has also shown that participating in SHGs can empower members, increasing both their bargaining power within the household but also social networks and connections within the village (Kochar et al. 2022; Kumar et al. 2019). We show that, in conjunction with reservations, SHGs allow more candidates

2. Goyal (2025) documents a party-based analogue in Delhi’s local elections, where quotas advance women’s careers through party organizations; in Bihar’s non-partisan GP elections, the SHG channel operates without parties to recruit, screen, or finance candidates.

to run and potentially increase overall voter turnout.

II. Institutional Background, Data, and Sample Selection

A. Self Help Group (SHG) program in Bihar

Bihar is among India’s poorest states, with over 120 million residents and persistently low levels of income and human development. In Bihar, the SHG program— called *Jeevika*—was piloted in 8 districts between 2007 and 2011 and scaled up statewide thereafter. Bihar currently includes over 12 million members as part of just over 1 million SHGs.³

The SHG model mobilizes 10–15 women from poor households, particularly from SC and ST groups, into savings and credit groups. These groups pool savings and access public funds: an initial grant of INR 15,000, followed by grants up to INR 50,000, contingent on regular meetings, savings, and repayment performance (Kochar et al. 2022; Datta et al. 2018). These funds are a critical source of low-cost credit in a context where formal financial access remains limited.

SHGs also offer platforms for leadership and collective action. Over time, groups are integrated into a federated structure—Village Organizations (VOs) and Cluster Level Federations (CLFs)—through which select members take on governance roles. These positions expose women to new skills in coordination and communication, while expanding their networks and influence within the community.

JEEViKA’s enrollment strategy disproportionately reached Bihar’s poorest households, who, in the state’s caste-segregated villages, are predominantly Dalit. The program entered villages through their hamlets, used social mapping to identify the poorest households, and mobilized these households first. Because hamlets are largely caste-homogeneous, poverty targeting also amounted in practice to caste targeting: the poorest Dalit women joined earliest, while upper-caste women were considerably harder to mobilize (Majumdar, Rao, and Sanyal 2026).⁴ Patterns of take-up from this experiment reflect this targeting. In treated GPs, SHG membership increased by 53.7 percentage

3. Numbers were accessed from the NRLM website in March 2024:
<https://nrlm.gov.in/shgOuterReports.do?methodName=showShgreport>

4. Majumdar, Rao, and Sanyal (2026) describe enrollment as focused on women from the most impoverished families, who are predominantly from historically marginalized castes, specifically the Scheduled Castes (SC) and note that the earliest joiners were mobilized quickly and enrolled early due to their “harsh lives,” while “upper-caste women proved challenging to integrate” (p. 1).

points among SC/ST households, compared with 43.7 percentage points among other households (Hoffmann et al. 2021).

B. Local Government and Political Reservation in Bihar

Bihar’s rural local governments are organized into Gram Panchayats (GPs), each serving around 12,000 people. Each GP is divided into an average of 13.6 wards. Voters directly elect two tiers of representatives: the Mukhiya (GP head) and ward members (WMs). Bihar has over 8,000 GPs and 100,000 elected WMs. Bihar’s GP elections are non-partisan by statute: candidates cannot contest on party tickets or symbols.

Reservations

History: Reservation status for GP-head seats was assigned in 2006 and held fixed for the 2006 and 2011 elections; seats were then reassigned under revised rules for the 2016 election. Seats for GP heads and WMs are reserved for SCs, STs and Extremely Backward Classes (EBCs) in proportion to their population share. Within each caste category — including open seats — up to 50% are further reserved for women. The first reservation cycle covered 2006–2016, and second covers 2016–2026. We obtained the lists of reserved seats, the GP population figures used to assign them, and the assignment rule itself from officials at the Bihar State Election Commission and the Department of Panchayati Raj; Kumar and Sharan (2025) validates the lists and the rule and describes the algorithm in detail, which we reproduce in Appendix Section XIII.

Entry in the Absence of Reservations is Rare: In the experimental sample, SC candidates have virtually no electoral viability in caste-unreserved GP head seats: SCs contest these seats — 94 SC candidates run in them — but none wins: no SC candidate takes any of the 144 caste-unreserved seats for which we observe a winner, in either arm, and fewer than 10% of candidates in control GPs are SCs, irrespective of women’s reservation.

Reservation Algorithm We exploit SC reservation status of GPs in Bihar during the 2016 local elections. Within each block, GP head seats are reserved for SCs based on the GP’s SC population. This creates a block-specific population threshold: GPs above the cutoff are typically reserved, while those below are not. We follow Kumar and Sharan (2025) and construct a running variable based on SC population. Further details are provided in Appendix Section XIII.

C. Data Sources and Samples

We draw on multiple data sources:

SHG Program Data We use two sources:

- *Administrative data* from Jeevika, covering all 8,392 GPs from 2007–2022. It includes group formation dates, member counts, and caste composition.
- *Scraped data* from the National Rural Livelihoods Mission (NRLM) website, covering 9.9 million SHG members. This includes names, caste, GP location, and father/husband names, and names of leaders (President and Secretary) of VOs/CLFs. Appendix Figure A1 shows program-expansion over time.

Political Entry Data We use publicly available data from the 2016 GP elections, covering 7,792 of 8,392 GPs. This includes candidate names, caste, gender, education, age, votes received, and winner. Reservation (for SC, ST, OBC, and women) is also included. We use data from the 2006 elections for control variables and 2021 elections to examine persistence. Ward-level election data is available for 109,352 wards across 8,093 GPs. Summary statistics are in Appendix Table A1.

GP-Level Characteristics We supplement the above with:

- *2011 Population Census*: We create control variables that we add to our main regressions.
- *2012 Socio-Economic and Caste Census (SECC)* data: we match political candidates to SECC and back out (a) household wealth and (b) sub-caste size of candidates. We are able to find about 61% of our candidates. Household wealth is based on PCA asset score of: land; the type of roof (concrete or not) and walls (whether made of burnt brick or concrete) of the main dwelling room; whether the house has four or more rooms; whether the household has a phone; and whether the household owns a vehicle. We proxy for sub-caste by surname size.⁵

5. In rural Bihar, surnames are more strongly linked to sub-castes than in many other regions. To address ambiguity, we cross-check paternal surnames and restrict matches to within broad caste groups (see Kumar and Sharan 2025). These steps help mitigate concerns about using surnames as a proxy for sub-caste identity.

- *Mission Antyodaya data*: This dataset - accessed via the SHRUG platform - provides us with downstream village infrastructure outcomes (Asher et al. 2021). Outcome variable descriptions in Table A2.

Samples Used

We rely on two main samples:

- **Experimental Sample**: Based on Hoffmann et al. (2021), this includes 89 GPs where the SHG program began in 2012 (treatment) and 90 GPs where it began in 2014 (control).
- **RDD Sample**: A larger sample based on the SC and female reservation thresholds. For SC reservations, we use block-level SC population cutoffs following Kumar and Sharan (2025). We construct an analogous running variable for female reservations. Appendix Figure A2 shows the resulting discontinuities, and Appendix Figure A3 shows that the number of SHGs formed by 2015 does not jump at either cutoff.

III. Results from the RCT

A. Empirical Strategy

To study the impact of exposure to SHGs on political entry, we first use the RCT sample to estimate:

$$Y_i = \beta_o + \beta_1 Treated_i + \mathbf{C}_i + pw_i + \epsilon_i \quad (1)$$

where Y_i is the number of SC/SC women candidates running for the GP head position in the 2016 elections. $Treated_i$ is an indicator variable for the earlier randomized introduction of the SHG program in 2012 (control group accesses SHGs in 2014). Controls, C_i , include: GP and SC population, area, distance to nearest town, and past gender reservation. The original RCT design randomized within 85 pair-wise strata, pw_i .⁶

Next, we use the additional variation in the formation of groups (num_shg_i) by 2015, and instrument the number of groups formed with ($Treated_i$); first-stage estimates are

6. Section VIII discusses how our experiment is well-powered to see reasonably small treatment effects.

reported in Appendix Table A3.

$$num_shg_i = \alpha_0 + \alpha_1 Treated_i + \mathbf{C}_i + pw_i + \nu_i \quad (2)$$

$$Y_i = \beta_0 + \beta_1 \widehat{num_shg}_i + \mathbf{C}_i + pw_i + \epsilon_i \quad (3)$$

Our next specification estimates separately the interaction effect of SHGs and reservations, R_i (for SCs):

$$Y_i = \beta_o + \beta_1 R_i + \beta_2 Treated_i + \beta_3 R_i Treated_i + \mathbf{C}_i + b_i + \epsilon_i \quad (4)$$

In this specification we use block fixed effects b_i since the RCT treatment and control samples were also balanced within blocks by design but using the pairwise strata would be too restrictive for including reservations (and interactions with reservations). Additionally, we show results using a version of Eq. (4) that instruments for reservation status using the SC population of the GP being above a threshold. We refer to this design as an RD-RCT design. Below, we describe the fuzzy RD design.

$$R_i = \alpha_0 + \alpha_1 1(Pop_i > T_b) + \alpha_2 (Pop_i - T_b) + \alpha_3 (Pop_i - T_b) * 1(Pop_i \geq T_b) + C_i + b_i + \eta_i \quad (5)$$

$$Y_i = \beta_o + \beta_1 R_i + \beta_2 Treated_i + \beta_3 R_i Treated_i + \mathbf{C}_i + b_i + \epsilon_i \quad (6)$$

Here T_b is the population threshold above which GPs are reserved for SCs (see Appendix Section XIII for a description of the reservation rule).

B. Threats to validity

Both compliance with the SHG program and balance of covariates in this experimental sample are robust and discussed in Hoffmann et al. (2021). If treatment affects outcomes independent of the number of groups formed, then the exclusion restriction for measuring the impact of the number of groups could be violated: early rollout may have changed the dynamics of women’s collective action beyond the groups it created, given groups more time to mature, altered the climate around women’s participation in public life and the support available to SHG-affiliated candidates, or changed which households join. We still use the instrument because it allows us to scale the experimental effect by the number

of groups the program created, which helps interpret its magnitude; we therefore read the per-group coefficient as the effect of early rollout expressed per additional group, rather than as the effect of forming one more group. Appendix Figure A4 plots cumulative group formation in the two arms: treated GPs form groups two years earlier, and the difference in the number of groups persists through 2020. Treated GPs also formed groups faster early on, adding about 30 more groups in their first two program years than control GPs did in theirs, so treatment varies the pace of early formation as well as the timing of access.

C. Effect on Candidate Pool

We first present the relationship between SHGs and political entry following Eq. (1). Panel A of Figure 1 shows that treatment effects on overall entry and on women’s entry are small and statistically insignificant (Appendix Tables A4 and A5). In contrast, consistent with SHGs targeting poor and marginalized castes, we find large and significant effects on the entry of SCs (2.2 additional candidates, 79%, $p = 0.042$) and SC women (1.2 additional candidates, 127%, $p = 0.057$).⁷ Appendix Table A7 instruments for the number of groups formed by treatment, following Eqs. (2) and (3), and shows that one additional SC (SC woman) candidate emerges for every 22.2 (38.5) SHGs formed. Given the small RCT sample, we also compute randomization-inference p -values. These are 0.04 and 0.064 for SC and SC women candidates, respectively, consistent with the conventional estimates.

Panel B of Figure 1 investigates where the treatment effect on SC entry comes from. In treated GPs without SC reservation, there is no increase in the entry of either SC or SC women candidates: the coefficients are small and statistically insignificant. Instead, almost all of the increase in entry is concentrated in SC-reserved GPs. Relative to unreserved GPs, treatment leads to between 3.7 ($p = 0.088$) and 4.4 ($p = 0.056$) additional SC women candidates in SC-reserved GPs, with the larger estimate instrumenting for reservation using the fuzzy-RD specification in Eqs. (5) and (6).⁸

Together, these results point to a clear pattern: longer exposure to SHGs increases the political entry of SCs, but only when combined with reservation.⁹ We find no correspond-

7. Our results are robust to pooling SCs with Scheduled Tribes (STs), who are very few in our sample (Appendix Table A6).

8. In Table A8, Col 1, we show that nearly 4.6 additional SC women candidates run in treated GPs reserved for both SCs and women, and the result is statistically significant. We report these results in the Appendix because the number of GPs reserved for both SCs and women in the RCT sample is small.

9. In Appendix Section XI, we provide evidence on the external validity of these findings using an alternative event-study design that exploits the timing of SHG entry into GPs across all of Bihar.

ing increase among women or non-SCs. Although these groups are also mobilized through SHGs, they are not the primary target group, are less poor on average, and do not benefit from a reservation policy that similarly narrows the pool of eligible candidates.¹⁰ Moreover, consistent with the idea that ward-level posts offer more limited benefits, we find no effects on political entry in ward elections (Appendix Table A9).¹¹

D. Who Enters?

Reservation tells us where the additional entry occurs; it does not tell us who the additional entrants are. We next turn to their characteristics. Using candidate affidavits and SECC data, we construct three measures of candidate type for each GP: (i) average years of education, (ii) average asset wealth,¹² and (iii) average sub-caste size, proxied by the prevalence of candidates' surnames in the SECC.¹³ We combine these measures into an electability index, weighting each component according to its ability to predict victory among candidates in control GPs.¹⁴

Figure 2 shows no effect on the composition of the candidate pool across all treatment GPs (full results in Appendix Tables A11 and A12). In *SHG + Reserved* GPs, however, candidates score lower on the electability index. This pattern is driven by lower education and asset wealth among candidates in these GPs. Thus, in the same reserved GPs where the increase in political entry is concentrated, the additional entry comes from candidates who are, *ex ante*, less socioeconomically advantaged and less electorally competitive.

E. Entry Among SHG Members and SHG Leaders

The new entrants are observably less electable. The natural question is whether they are drawn from the SHGs themselves – as members, or as leaders.

When we compare treatment and control GPs (Table 1) and look at backgrounds of

10. This is true even for the marginalized Extremely Backward Classes (EBCs), who are both poor and have reservation in place for them. Two explanations are plausible. First, EBCs are roughly twice as large a group as SCs in Bihar and are wealthier on average. Second, they are a sufficiently strong political group to win even without reservation: EBC candidates win 14% of caste-unreserved seats in our sample, while SC candidates win none.

11. Appendix Figure A5, however, shows rising SC female participation in wards with higher baseline SC populations, consistent with the framework's corollary.

12. We match 1,287 of 2,108 candidates (61%) across 172 of 179 GPs to households in the 2011–12 SECC. Matched candidates are more likely to be SC, marginally more educated, and slightly older (Table A10). Appendix IX provides details.

13. That is, the number of GP residents in the SECC who share the candidate's surname, cross-checked within broad caste group; we use its log in the analysis.

14. Appendix XII provides details on the construction of the index.

contesting female candidates (Panel A), a striking pattern emerges. Among 991 female candidates across 137 GPs, the SHG match rate is 12.6% in treatment GPs versus 6.3% in control GPs (Col 1). This further rises to 24.4% in *SHG + Reserved* GPs (Col 2). For SHG leaders, the match rate rises by 2.2 percentage points in treatment GPs, from zero in control GPs (Table 1, Cols 3 and 4). These results highlight that political entry among SHG-linked women is significantly more likely in treatment areas, especially in *SHG + Reserved* GPs. Moreover, in the RCT sample but also in the universe of all GPs, SHG leaders exhibit a markedly higher propensity to contest elections compared to members (Tables A13, A14).¹⁵

We caution against interpreting these results as implying that most female candidates in 2016 were not SHG members. Our conservative name-matching algorithm — requiring high similarity thresholds across datasets with inconsistent spellings — likely understates true overlap.

A natural concern is that these match-rate gaps are mechanical: treated GPs have more SHG members, so more candidates would match to the rolls even if members were no likelier to contest. Appendix Table A15 addresses this by benchmarking each match rate against SHG prevalence among the GP’s women. If candidates were a random draw from the female population, the match rate would rise in proportion to prevalence, and the ratio of the two would be unchanged by treatment. For members, prevalence rises by 59% while the match rate more than doubles; the over-representation ratio rises from 0.36 to 0.67, and the increase is borderline significant (Col 3, $p = 0.10$). For leaders the benchmark fails decisively: growth in the number of leaders can account for a rise of at most 0.2 percentage points in the leader match rate, an order of magnitude less than the observed increase, and the over-representation ratio rises by 13.3 ($p = 0.051$). SHG-linked candidacy – leader candidacy in particular – rises well beyond what growth of the rolls alone would produce.

Overall, SHG participation, especially in reserved GPs, facilitates political entry among women from poorer and less privileged backgrounds. While SHG membership accounts for much of this effect, leadership positions appear to provide an additional springboard into formal politics. Yet, in this context, entry is rarely an individual act: it often reflects household-level decision-making. We therefore next turn to the household as the unit of analysis.

15. In the overall sample of GPs, the matches are low in absolute terms, but show meaningful patterns. Just 0.08% of SHG members and 0.65% of SHG leaders (Presidents and Secretaries of Village and Cluster-Level Federations) appear in the candidate list (see Table A13). Nonetheless, since SHG leaders are over eight times more likely to run for office than regular members, this suggests that leadership roles within the SHG structure may serve as a pathway into formal politics. This may not be entirely causal since unobserved factors could drive both SHG leadership and political entry.

F. Entry Among SHG Member and Leader *Households*

We match both the candidates and the SHG dataset to households within the SECC. Thus, for each candidate, we can identify the households they come from and whether that household had an SHG member or leader.

Figure A6 shows that SHG households field both male and female candidates—if anything, more men than women. This supports our argument that SHG benefits accrue at the household level, a finding robust to varying the matching threshold.

Panel B of Table 1 compares treated and control GPs. In control areas, 32.8% of male and 39.6% of female candidates come from SHG member households; in treated GPs, these rise by 8.6 and 11.5 percentage points, respectively. Cols 3–4 show that effects are concentrated in *SHG + Reserved* GPs.

Panel C of Table 1 reports similar patterns for SHG leader households: 4.4% of male and 6.1% of female candidates in control GPs come from such households, roughly doubling in treated GPs (Cols 1–2). Reservation does not further raise entry from leader households, likely because these households were inframarginal in reserved GPs but marginal in unreserved ones.

G. Turnout

If SHG households supply the candidates, SHG networks may also mobilize votes. We next examine turnout. Anecdotally, there is evidence – Sanyal, Rao, and Majumdar (2015) quote a Jeevika woman: “Women came in large numbers from eight different villages for the election. And everyone voted for me. You know why? Because I’ve made groups everywhere.” We measure turnout by the number of votes polled in GP elections, controlling for the population of GPs. An increase in turnout could reflect SHG women leveraging their expanded networks to mobilize voters more effectively.

We report four findings in Appendix Table A16 that strongly suggest that SHGs increase turnout, but only for SC candidates when seats are reserved: first, treatment does not increase voter turnout overall (Col 1); second, in *SHG + Reserved* GPs, turnout increases by 414 votes or 9.3% relative to unreserved GPs (Col 3, $p = 0.082$); third, votes polled for SC candidates rise by 397 votes in *SHG + Reserved* GPs relative to unreserved GPs (Panel B, Col 3, $p = 0.072$); finally, votes per SC candidate does not move (Col 5), suggesting that the increase in votes comes from new candidates being able to mobilize newer voters.

H. Effect on Winners

More candidates and more votes need not change electoral outcomes. We now ask whether they change who wins. Our previous results provide some indication of where and whether a new type of winner may emerge: SCs enter more. However two findings suggest that entry may not convert to winning: first, the marginal entrant is less electable. Moreover, the average GP has 12.1 candidates, suggesting a win chance on a random draw to be 8%. Less electable candidates have even lower probability of winning.

Consistent with this, we find no change in the caste/gender identity of winners in unreserved seats. Panel A of Figure 3 shows that the winner is no more likely to be a woman in a gender unreserved seat, no SC wins in an open seat and no SC-woman is more likely to win in SC-open seats. Moreover, as Panel B shows, winners are no more likely to be SHG members or leaders. Turning to winner characteristics, we do not see any effects on education, asset scores, sub-caste size or electability of the winning candidates (Appendix Tables A17 and A18).¹⁶

Overall, our results on winners are consistent with our entry findings and echo other work studying SHGs (Majumdar, Rao, and Sanyal 2026). For instance, Sanyal, Rao, and Majumdar (2015) quote a woman SHG-leader in Bihar who contested and lost the GP election: “I had fought for [the GP Head] elections, but I lost. The other candidate bought the seat; he spent thirteen lakhs. And I had no money.”

I. Downstream Outcomes

Despite having no effects on winners, the candidacy of SCs and turnout suggests a shifting of the local political equilibrium that could influence downstream outcomes. We focus on entry in subsequent elections, public goods and SHG performance 12 years later (see Figure 4).¹⁷ Political effects persisted beyond the 2016 elections: in treated GPs, more SC women continued to contest (Panel A of Figure 4; Table A20). Winning also begins to move: in the twenty seats reserved for SCs but not for women, six of ten treated GPs elected an SC woman in 2021, against one of ten in control (Fisher exact $p = 0.057$; Appendix Table A21; Figure A7). This likely reflects higher SHG presence and better functioning groups through 2024–25 (see bottom 3 estimates in Figure 4).

However, this sustained entry did not translate into changes in public goods or welfare

16. The effective number of candidates and the margin of victory are also unchanged (Appendix Table A19).

17. We describe how outcomes are constructed in Table A2.

program outcomes (Figure 4; Tables A24, A25), even in *SHG + Reserved* GPs (Panel B of Figure 4). Besides the fact that this could be shaped by the absence of a new class of winners, it is also possible that reservation alone was already effective in improving SC citizens’ welfare: evidence suggests that SC GP heads prioritize SC welfare even during a time when the SHG movement was still in its infancy (Kumar and Sharan 2025). Finally, it is possible that SHG exposure improves outcomes, but because control GPs also accessed SHGs within two years of the intervention, we do not find differential effects in this instance.

IV. A Simple Framework

To rationalize our findings, we develop a simple model of political entry as a household-level decision, drawing on the valence-based framework of Dal Bó and Finan (2018); we treat the household as the deciding unit because, in this setting, candidacy is a family investment and SHG benefits accrue to the member’s whole household (Appendix X; Prillaman 2023b; Bamezai 2025). A household’s valence – its electability – governs its probability of winning a seat. The household fields an eligible member when the value of the office, weighted by that probability, plus a small non-office benefit of standing (status, visibility, service), covers the cost of contesting:

$$P(v; \theta) B + b \geq C.$$

Two interventions shift this margin. SHG exposure raises P for member households (wider networks, public visibility, demonstrated leadership) and eases C through savings and credit.¹⁸ Caste reservation (θ) restricts the field to SC rivals, discretely raising P ; a gender quota, by contrast, restricts which member a household may field but leaves the caste composition of the field unchanged.

Two features of the setting discipline the framework’s predictions. First, in caste-unreserved GP-head seats SC candidates have low electoral viability. This is backed by our data: 94 SC candidates contest caste-unreserved seats, and none of the 144 such seats for which we observe a winner elects an SC candidate, in either arm; fewer than 10% of control-GP candidates are SC, with or without a women’s quota. Second, in 2016 the GP headship and the ward seat differed starkly in what the office was worth: the GP head controlled funds, works, and appointments, while ward members had virtually no authority or fi-

18. Section G provides direct evidence for the mobilization channel: treatment raises votes polled for SC candidates only in SC-reserved GPs (with an RD-RCT estimate similar in magnitude), while votes per candidate are unchanged – members appear to mobilize additional voters for candidates of their own type rather than dividing a fixed pool.

nancial powers (Sharan 2021). Ward seats are also cheap to contest: a ward averages roughly 900 residents, against roughly 10,000 for a GP.

The framework then yields four predictions, derived formally in Appendix X. *(i)* SHG exposure does not change SC candidacy in caste-unreserved head seats – whether or not the seat carries a gender quota – and produces no SC winners there (Sections C and H). *(ii)* In SC-reserved seats, SHG exposure raises SC candidacy, and where a women’s quota compounds the caste quota the fielded member is a woman (Section C). *(iii)* The entrants drawn in are marginal: the valence of the candidate pool falls where entry rises (Section D). *(iv)* Entry does not translate into victory: induced entrants have lower winning odds than every household that would have contested anyway, so the identity and characteristics of winners barely move (Proposition 1; Section H). At the ward level, where the office was worth little in 2016, entry does not respond at all even though seats fill (Proposition 2; Appendix Table A9).

V. Conclusion

SHGs primarily aim to improve women’s financial security, but a growing body of research demonstrates their broader impact on communities (Díaz-Martín et al. 2022). For example, Prillaman (2023b) shows how an NGO-led SHG movement in Madhya Pradesh fosters women’s collective action, having transformative effects on their non-electoral political participation. Similarly, Sanyal, Rao, and Majumdar (2015), studying the initial rollout of Jeevika, suggest that SHGs can create a “mini-social movement” that challenges traditional caste structures and patriarchy within villages. Building on this literature, this paper shows that SHGs can also serve as an informal institutional complement to formal affirmative action: SHG presence increases political entry of marginalized groups.

Our framework suggests three explanations. First, SHG formation may help SC households bear the costs of contesting elections, enabling poorer and less educated candidates to enter the fray. While we lack data on individual loans, Hoffmann et al. (2021) estimate the average loan per member to be around INR 4,000. Given that the official spending limit for GP head elections was INR 40,000 in 2016 (Press Trust of India 2016)—and actual spending is reported to be considerably higher (Chaudhary 2016)—the financial support from SHGs alone is unlikely to explain all the observed entry. Second, SHGs may provide a platform for developing leadership skills, creating a pathway into politics. Third, the program’s expanded social networks may have enhanced voter mobilization and turnout, further increasing electoral participation.

However, this increased entry does not change who wins in the elections or affect downstream outcomes. SHG-linked entrants have relatively low electability and enter an already crowded field. This does not mean that their entry is inconsequential. In deeply hierarchical settings, the entry of historically excluded women and men into electoral politics may itself carry symbolic value and requires them to undertake a costly political action. Moreover, our estimates capture the effect of only two additional years of SHG presence. More sustained exposure may eventually increase candidates' prospects of victory and, through electoral success, affect downstream outcomes. We see some evidence of this with more SC women winning in the long-run, driven primarily by SC-reserved seats. More broadly, the rise of women and SCs in India's local politics, achieved largely through reservation (Chattopadhyay and Duflo 2004; Kumar and Sharan 2025), has coincided with the spread of women's SHGs, which now reach a majority of rural households (Díaz-Martín et al. 2022). Our results suggest that the second contributed to the first, but only in combination with reservation: on their own, SHGs did not change who contests. Future research should examine whether such effects emerge over longer time horizons and in other contexts.

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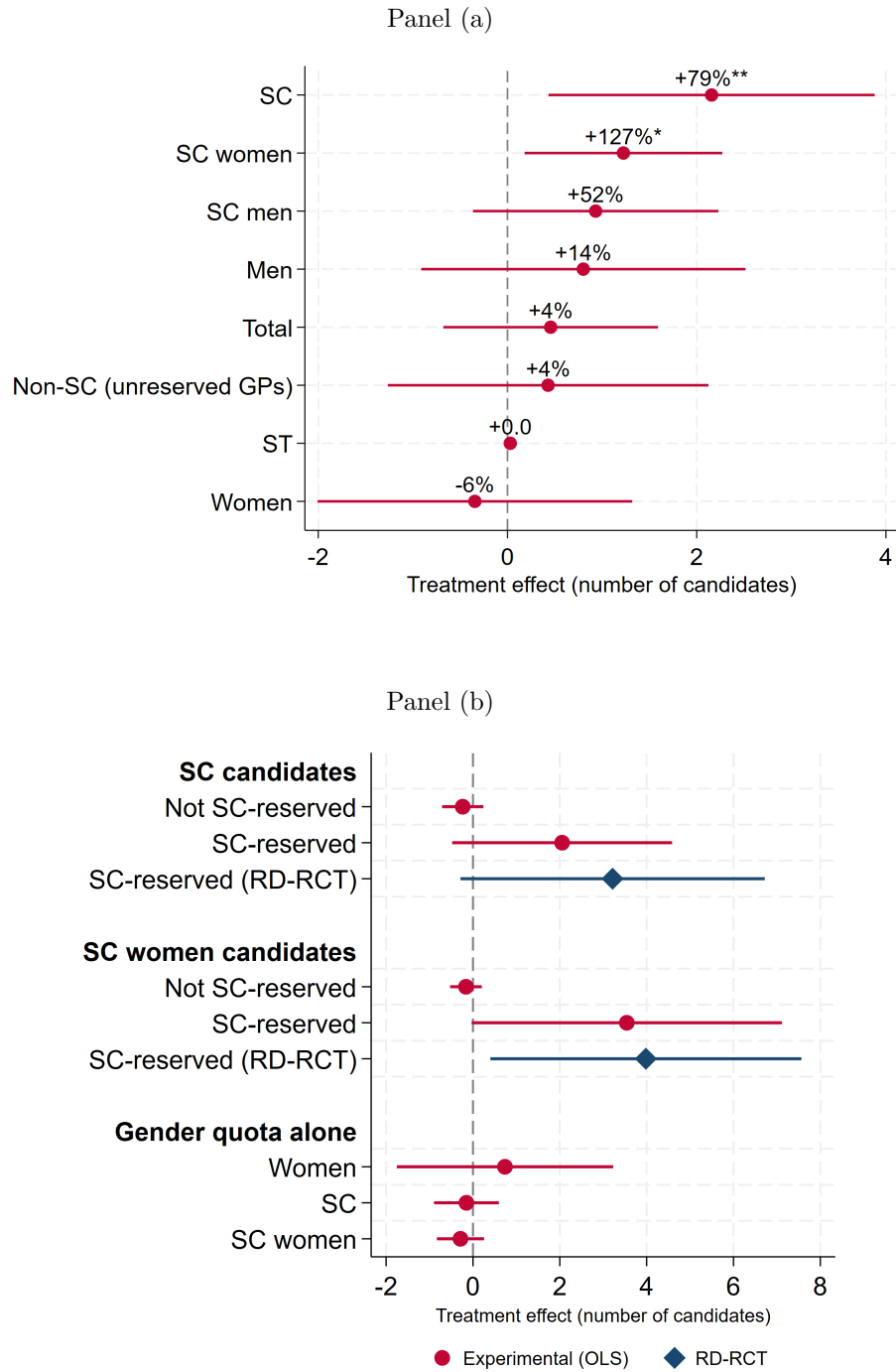
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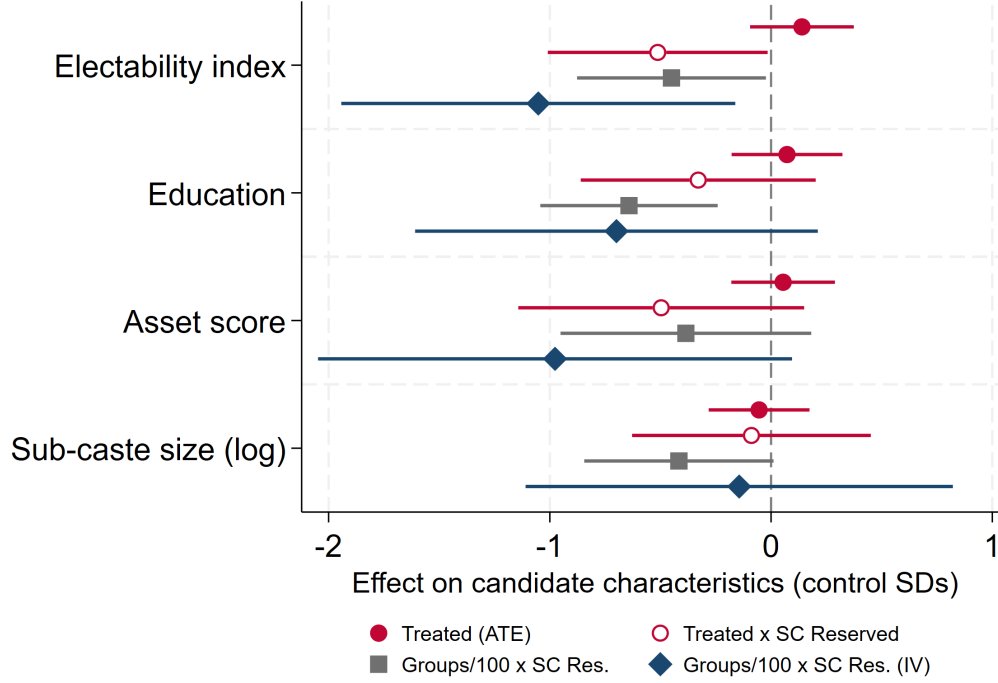
VI. Main Tables and Figures

Figure 1: Who Enters



Notes: The sample is the 179 RCT GPs; outcomes are the number of candidates of each type contesting the 2016 GP-head election, and Treatment is an indicator for early-treatment GPs. Panel (a) plots the treatment effect for each candidate type, from regressions with randomization-strata fixed effects; rows are sorted by effect size, and labels express each effect as a percentage of the control-group mean (the ST row reports the effect in levels, as its control mean is near zero). Panel (b) splits the SC and SC women effects by the GP's 2016 SC-reservation status. Circles are OLS estimates from regressions interacting treatment with SC reservation, with block fixed effects: the "Not SC-reserved" rows show the treatment effect in unreserved GPs, and the "SC-reserved" rows the total effect in reserved GPs (main effect plus interaction). Diamonds (RD-RCT) instrument reservation status and its interaction with treatment using whether the GP's running variable lies above its block's reservation threshold, as in Eqs. (5) and (6), including the running variable as a control and using a small-sample correction. The bottom block reports the interaction of treatment with the 2016 women's reservation, estimated among caste-unreserved GPs. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and robust standard errors; bars are 90% confidence intervals; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$ on the labels. Full regression results appear in Tables A7 and A5.

Figure 2: Candidate Characteristics



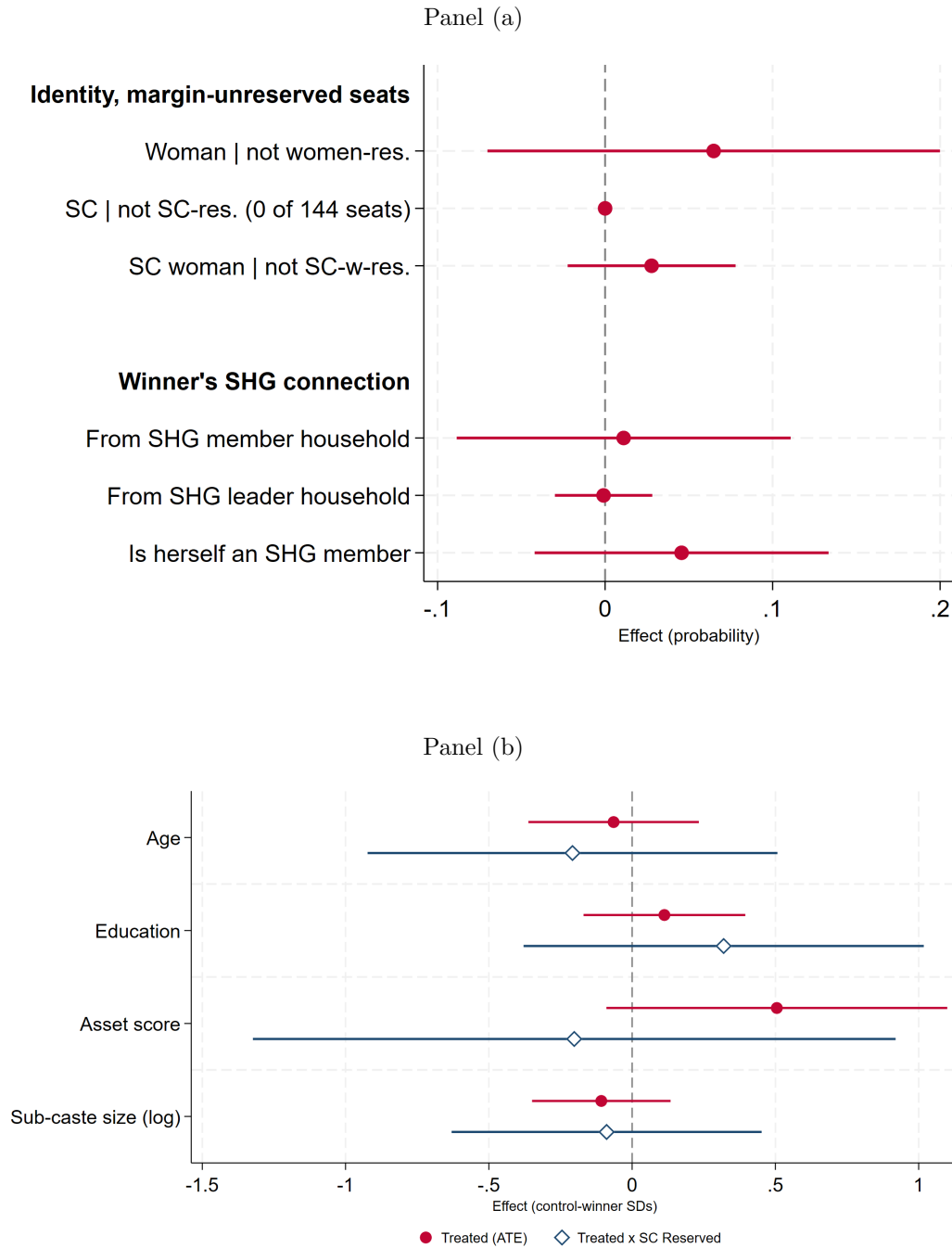
Notes: Each outcome is the GP-level average of a characteristic of the candidates matched to the SECC, expressed in control-group standard deviations: years of education, asset score, log sub-caste size (proxied by surname prevalence in the SECC), and the electability index — the weighted combination of these three components, with weights estimated from a regression of an indicator for winning on the standardized components among candidates in control GPs (with GP fixed effects). Filled circles show the treatment effect across all GPs, where Treatment is an indicator for early-treatment GPs; hollow circles show the interaction of treatment with the GP’s 2016 SC-reservation status. Squares replace the treatment indicator with the number of SHG groups formed by 2015 (in hundreds) interacted with SC reservation; diamonds instrument the number of groups and its interaction using treatment assignment and its interaction with reservation status. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and block fixed effects, with robust standard errors; bars are 90% confidence intervals. Full regression results appear in Tables [A11](#) and [A12](#).

Table 1: Political Entry from SHG Households

Panel A: Female candidates matched to SHG members and leaders (rates x 100)				
	(1)	(2)	(3)	(4)
Treated	6.279*** (2.335)	0.122 (3.032)	2.160*** (0.558)	1.270* (0.757)
SC_Reserved		-3.991 (2.505)		0.070 (0.656)
Treated X SC Reserved		19.014** (7.564)		2.313 (1.913)
Observations	982	982	991	991
Sample	Members	Members	Leaders	Leaders
Control Mean	6.31	6.31	0.00	0.00
Panel B: Candidate households containing an SHG member (rates x 100)				
	(1)	(2)	(3)	(4)
Treated	8.940* (5.222)	11.496* (6.310)	4.810 (5.284)	9.128 (7.703)
SC_Reserved			-8.990 (7.520)	-29.166*** (10.236)
Treated X SC Reserved			16.858 (10.481)	22.986 (14.873)
Observations	627	459	627	459
Sample	Male Cand HH	Female Cand HH	Male Cand HH	Female Cand HH
Control Mean	32.8	39.6	32.8	39.6
Panel C: Candidate households containing an SHG leader (rates x 100)				
	(1)	(2)	(3)	(4)
Treated	4.628** (1.859)	6.071** (3.010)	3.931* (2.137)	6.911** (2.947)
SC_Reserved			-1.663 (2.999)	-2.587 (5.986)
Treated X SC Reserved			2.766 (3.910)	-0.971 (6.734)
Observations	740	543	740	543
Sample	Male Cand HH	Fem Cand HH	Male Cand HH	Fem Cand HH
Control Mean	4.4	6.1	4.4	6.1

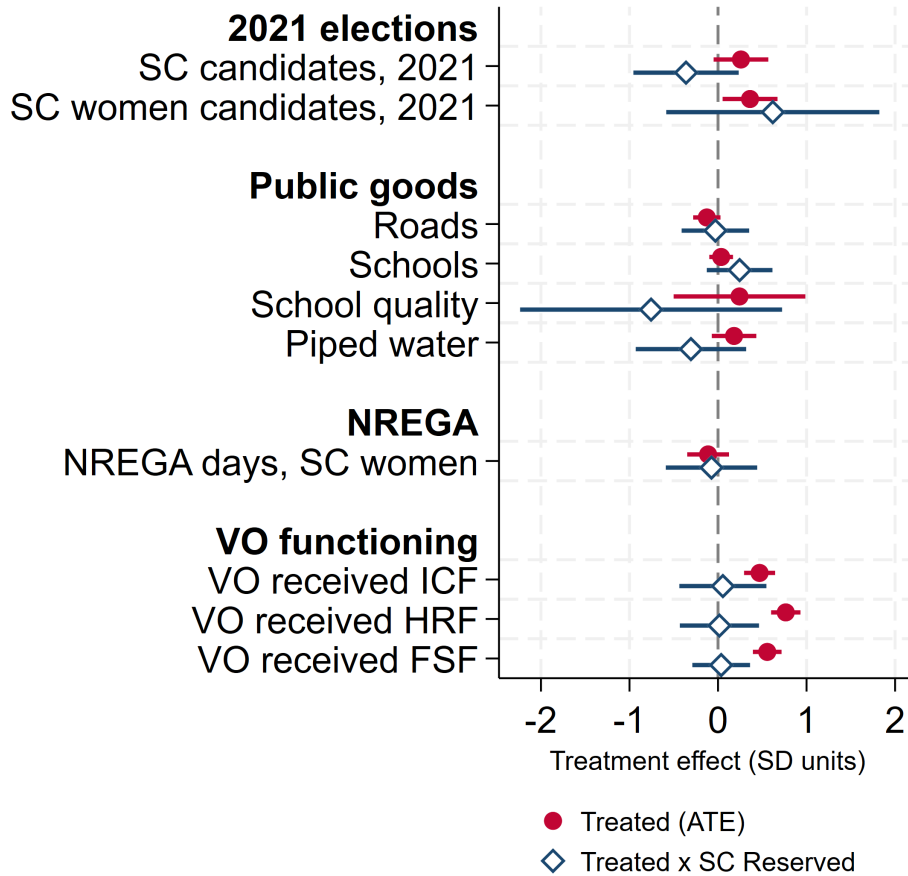
Notes: All outcomes are match indicators multiplied by 100; *Treated* is an indicator for early-treatment GPs. Panel A is at the candidate level, for female candidates in the RCT sample: the outcome is whether the candidate is herself matched by name to an SHG member (columns 1–2; control mean 6.3) or to an SHG leader (Cols 3–4; control mean 0.0), using the match-score threshold of 0.98 and groups formed by 2015. Panels B and C are at the candidate-household level, for candidates matched to SECC households: the outcome is whether the candidate’s household contains an SHG member (Panel B) or an SHG leader (Panel C), estimated separately for male-candidate households (Cols 1 and 3) and female-candidate households (Cols 2 and 4). Cols 2 and 4 of Panel A, and Cols 3 and 4 of Panels B and C, add the GP’s 2016 SC-reservation status and its interaction with treatment. We use a more conservative household-level threshold of 0.99 in Panels B and C. Panel A regressions include baseline GP controls; all regressions include block fixed effects, with standard errors clustered at the GP level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Appendix IX details the matching procedure.

Figure 3: Who Wins



Notes: Panel (a) plots treatment effects on the identity of the 2016 GP-head winner and on the winner's SHG connection, in probability units; Treatment is an indicator for early-treatment GPs. The first block restricts to seats where the identity in question is not guaranteed by a quota: the probability that the winner is a woman, among GPs not reserved for women ($N = 101$); the probability that the winner is SC, among caste-unreserved GPs — plotted at zero with no confidence interval, because no SC candidate wins any of the 144 caste-unreserved seats for which we observe a winner, in either arm; and the probability that the winner is an SC woman, among GPs not reserved for both SCs and women ($N = 165$). These regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and block fixed effects. The second block uses the SECC-SHG match ($N = 168$ GPs): indicators for whether the GP's winner comes from an SHG member household, from an SHG leader household (VO office-holders), and for whether the winner is herself a matched SHG member; matches use the conservative name-match threshold (score ≥ 0.99) and groups formed by 2015 (Appendix IX), and these regressions include block fixed effects. Panel (b) plots effects on the winner's characteristics — age, years of education, SECC asset score, and log sub-caste size (proxied by surname prevalence) — standardized against control-group winners. Filled circles are treatment effects with randomization-strata fixed effects; hollow diamonds are the interaction of treatment with 2016 SC-reservation status, with block fixed effects; both include the GP controls above. Characteristics are observed for 176 (age), 174 (education), 139 (sub-caste), and 88 (asset score; SECC-matched winners) GPs. Robust standard errors; bars are 90% confidence intervals. Full regression results appear in Tables A17 and A18.

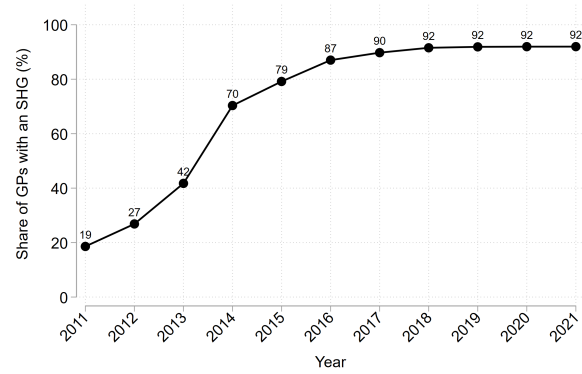
Figure 4: Downstream Outcomes



Notes: Each row is a standardized downstream outcome: the number of SC and SC women candidates in the 2021 GP elections; the presence and coverage of roads, schools, school quality, and piped water by 2019; NREGA days worked by SC women over 2017–2020; and the share of the GP's village organizations (VOs) that had received each of three SHG program funds (ICF, HRF, FSF) by 2024. Filled circles show the treatment effect, from regressions with strata fixed effects; hollow diamonds show the interaction of treatment with the GP's 2016 SC-reservation status, from regressions with block fixed effects that include the main effects. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and robust standard errors; bars are 90% confidence intervals. Winner characteristics appear in Figure 3 and Tables A17 and A18.

VII. ONLINE APPENDIX

Figure A1: Percentage of villages reached by the SHG program (2011-2021)



Notes: Total GPs in Bihar state = 8,392 (Census 2011). Data source: SHG program administrative data.

Table A1: Number of female and SC candidates running for local elections in 2016

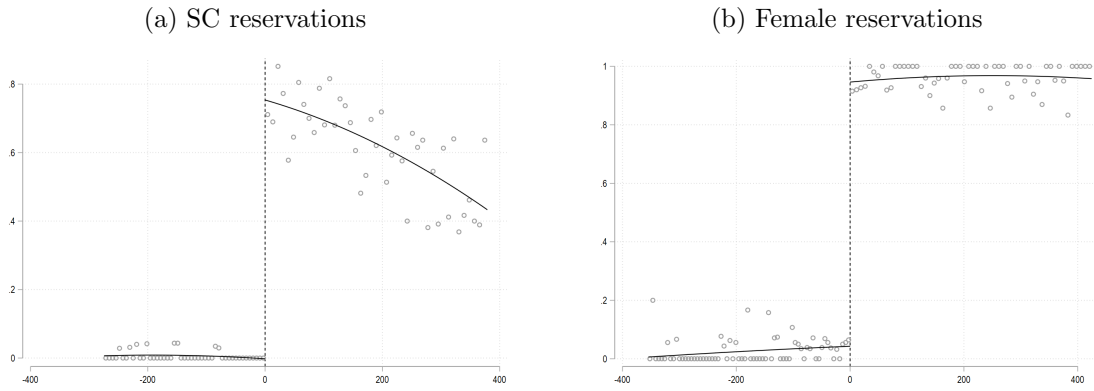
	Num female candidates	Num SC candidates	Num GPs
Reserved for women in 2016	12.1 (5.6)	2.8 (5.4)	3,720
Reserved for SC in 2016	6.9 (7.1)	14.8 (5.1)	1,377
Reserved for OBC in 2016	6.4 (7.4)	0.0 (0.2)	1,416
Unreserved in 2016	1.1 (1.6)	0.9 (1.6)	2,879

Notes: 557 and 573 GPs have both women + SC and women + OBC reservations respectively. Data source: 2016 GP Elections administrative data.

Table A2: Downstream Outcome Description

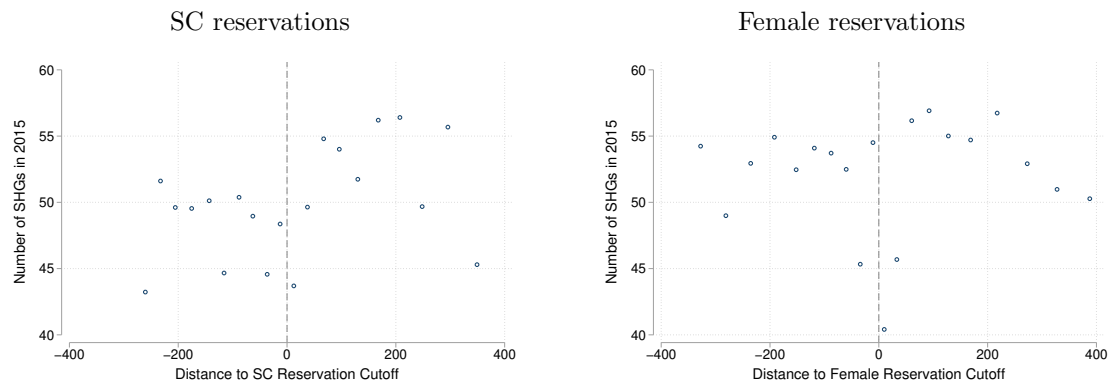
Variable		Description	Source	Year
All SCs		Number of SC candidates running in the 2021 local elections	Bihar Elections	2021
SC Women		Number of Women SC candidates running in the 2021 local elections	Bihar Elections	2021
Roads		Total roads available in GP from 2 variables that measure the extent to which the village is connected by all weather roads and there are internal gravel roads	Mission Antyodaya	2019
Schools		Total number of schools including primary, middle, high, and senior secondary schools	Mission Antyodaya	2019
School Quality		Total of if primary school has electricity, computer, playground, drinking water, mid day meal scheme, and number of teachers	Mission Antyodaya	2019
Piped Water		Share of GP connected with piped water	Mission Antyodaya	2019
NREGA workdays	SC	Number of workdays completed by SC individuals in GP between 2017-2020	NREGA MIS	2017-2020
Share of VOs received ICF		SHGs are eligible to receive an Initial Capitalization Fund (ICF) of \$625 after they have conducted meetings and save regularly for atleast 6 months	SHG MIS	2024
Share of VOs received HRF		The Health Risk Fund (HRF) is made available to SHG members to borrow from at low interest rates in the case of a health emergencies	SHG MIS	2024
Share of VOs received FSF		The Food Security Fund (FSF) is typically used to make purchases of subsidized food rations	SHG MIS	2024

Figure A2: Block level population based reservations



Notes: Panel (a) shows probability of GP a being reserved for SCs in the 2016 elections; the SC population of a GP has been normalized by the block-level threshold SC population. Panel (b) shows probability of GP a being reserved for women in the 2016 elections; the population of a GP has been normalized by the block-level threshold population. Both figures show a clear jump in the probability of reservation once the threshold is crossed.

Figure A3: Number of SHGs in 2015 around the SC and Female reservation cutoff



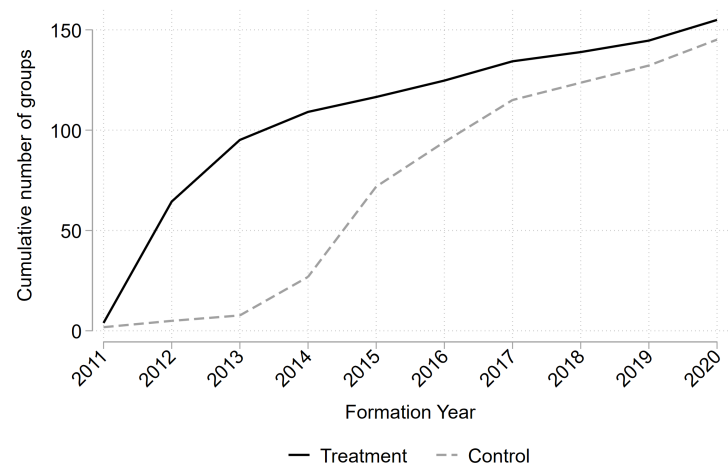
Notes: Figure shows average number of SHGs formed by 2015 above and below the SC and female reservation cutoffs.

Table A3: First Stage Regressions

	(1)	(2)
	# of SHGs	Years of exposure
Treated	47.963*** (6.527)	2.113*** (0.036)
Observations	179	179
Fixed Effects	Strata	Strata
Control Mean	71.86	0.86

Notes: The table shows the first stage for number of SHGs in Col 1 and years of exposure to the SHG program in Col 2. Treated is an indicator for earlier rollout of the SHG program. All regressions have strata fixed effects and robust standard errors in brackets.

Figure A4: Evolution of Groups in RCT sample



Notes: The figure shows the evolution of SHGs across treatment and control GPs. Eighty nine treatment GPs randomly assigned to receive the SHG program in 2012, 90 control GPs received program in 2014.

Table A4: Number of total, women, male, non-SC, and SC male candidates running in treatment GPs

	(1)	(2)	(3)	(4)	(5) SC
	Total	Women	Male	Non-SC	Male
Treated	0.456 (0.690)	-0.345 (1.011)	0.801 (1.042)	-1.701 (1.049)	0.932 (0.789)
Observations	179	179	179	179	179
Fixed Effects	Strata	Strata	Strata	Strata	Strata
Control Mean	12.14	6.22	5.92	9.40	1.78

Notes: The outcome variable is the number of candidates running for the GP head position in the 2016 elections, by candidate group: all candidates (Col 1), women (Col 2), men (Col 3), non-SC candidates (Col 4), and SC men (Col 5). Treated GPs are GPs where the SHG program was introduced in 2012; control GPs received the program in 2014. All regressions include strata fixed effects, GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status), and robust standard errors. Corresponding estimates for SC and SC women candidates appear in Table A7, and for SC and ST candidates pooled in Table A6. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A5: Number of Women candidates running in treatment GPs in the 2016 local elections

	(1)	(2)	(3)	(4)	(5)
	RCT	IV	Quota Het Women	Quota Het SC	Quota Het SC Women
Treated	-0.345 (1.011)		-0.062 (0.561)	-0.163 (0.330)	-0.072 (0.123)
# of Groups (instrumented)		-0.007 (0.021)			
Women Reserved			10.182*** (1.045)	-0.258 (0.329)	0.454** (0.209)
Treated X Women Reserved			0.737 (1.515)	-0.150 (0.455)	-0.288 (0.331)
Observations	179	179	147	147	147
Fixed Effects	Strata	Strata	Block	Block	Block
Sample	All	All	Caste-unres.	Caste-unres.	Caste-unres.
Control Mean	6.22	6.22	6.52	0.78	0.38

Notes: The outcome variable in Cols 1 and 2 is the number of women candidates running for the GP head position in the 2016 elections. Col 1 reports the treatment effect with strata fixed effects; Col 2 instruments the number of SHGs formed by 2015 with treatment assignment. Cols 3–5 interact treatment with the GP's 2016 women's reservation among caste-unreserved GPs, with block fixed effects and main effects shown, for the number of women (Col 3), SC (Col 4), and SC women (Col 5) candidates; these are the estimates plotted in the bottom block of Panel (b) of Figure 1. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A6: Number of SC and ST candidates running in treatment GPs in the 2016 local elections

Panel A: All SCs and STs			
	(1)	(2)	(3)
	RCT	IV	RCT Het
Treated	2.142** (1.040)		-0.215 (0.293)
# of Groups		0.044*** (0.017)	
SC_ST_Reserved			13.581*** (1.131)
Treated X SCST Reserved			2.264 (1.634)
Observations	179	179	179
Fixed Effects	Strata	Strata	Block
Control Mean	2.79	2.79	2.79
Panel B: SC ST Women			
	(1)	(2)	(3)
	RCT	IV	RCT Het
Treated	1.195* (0.613)		-0.138 (0.231)
# of Groups		0.025*** (0.009)	
SC_ST_Reserved			4.162*** (1.209)
Treated X SCST Reserved			3.547* (2.128)
Observations	179	179	179
Fixed Effects	Strata	Strata	Block
Control Mean	1.00	1.00	1.00

Notes: The outcome variable is the number of SC and ST candidates in Panel A and SC and ST Women candidates in Panel B. Col 1 shows the results for the RCT. Col 2 shows the number of groups in 2016 instrumented using the *Treated* instrument. Col 3 shows results using the SC or ST reservation indicator in 2016 and the interaction with earlier SHG rollout. All regressions include village controls including GP total and SC and ST population, area, and distance to the nearest town controls, as well as if the GP was reserved in the past for women. Robust standard errors are shown in brackets.

Table A7: Number of SC and SC Women candidates running in treatment GPs

Panel A: All SCs					
	(1)	(2)	(3)	(4)	(5)
	RCT	IV	RCT Unreserved	RCT Het	RD-RCT
Treated	2.157** (1.048)		-0.084 (0.306)	-0.236 (0.290)	-0.331 (0.332)
# of Groups (instrumented)		0.045* (0.024)			
SC Reserved				13.624*** (1.117)	13.914*** (2.332)
Treated X SC Reserved				2.289 (1.614)	
Treated X SC Reserved (RD-RCT)					3.546 (2.266)
Observations	179	179	147	179	179
Fixed Effects	Strata	Strata	Strata	Block	Block
Sample	All	All	Unreserved	All	All
Control Mean	2.74	2.74	0.78	2.74	2.74
Panel B: SC Women					
	(1)	(2)	(3)	(4)	(5)
	RCT	IV	RCT Unreserved	RCT Het	RD-RCT
Treated	1.225* (0.635)		-0.204 (0.137)	-0.161 (0.222)	-0.423 (0.283)
# of Groups (instrumented)		0.026* (0.014)			
SC Reserved				4.096*** (1.261)	-0.753 (1.747)
Treated X SC Reserved				3.704* (2.157)	
Treated X SC Reserved (RD-RCT)					4.405* (2.292)
Observations	179	179	147	179	179
Fixed Effects	Strata	Strata	Strata	Block	Block
Sample	All	All	Unreserved	All	All
Control Mean	0.97	0.97	0.38	0.97	0.97

Notes: Outcomes are the number of SC candidates (Panel A) and SC women candidates (Panel B) contesting the 2016 GP-head election in the 179 RCT GPs; in SC-reserved GPs every candidate is de jure SC, so counts there equal the size of the field, which the SC Reserved main effect largely reflects. *Treated* is an indicator for early-treatment GPs. Col 1 is the experimental estimate. Col 2 instruments the number of SHG groups formed by 2015 with treatment assignment, with small-sample standard errors. Col 3 estimates the experimental effect among the 147 GPs not reserved for SCs. Col 4 adds 2016 SC-reservation status and its interaction with treatment. Col 5 instruments reservation status and its interaction with treatment using whether the GP's running variable lies above its block's SC-population threshold, as in Eqs. (5) and (6), including the running variable as a control and with small-sample standard errors. Fixed effects as indicated; all regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A8: Number and Characteristics of Candidates running in SC and Female Reserved GPs in the 2016 local elections

	(1) SC Women	(2) Education	(3) Asset Score	(4) Jati Population
Treated	-0.075 (0.195)	0.168 (0.156)	0.132 (0.150)	-0.032 (0.149)
SC_Fem_Reserved	10.860*** (1.485)	-0.280 (0.604)	0.167 (0.765)	0.042 (0.564)
Treated X SC Fem Reserved	4.599** (2.208)	-0.764 (0.659)	-0.973 (0.817)	-0.259 (0.632)
Observations	179	179	175	179
Fixed Effects	Block	Block	Block	Block
Sample	All	All	All	All
Control Mean	2.74	6.47	0.72	6.80

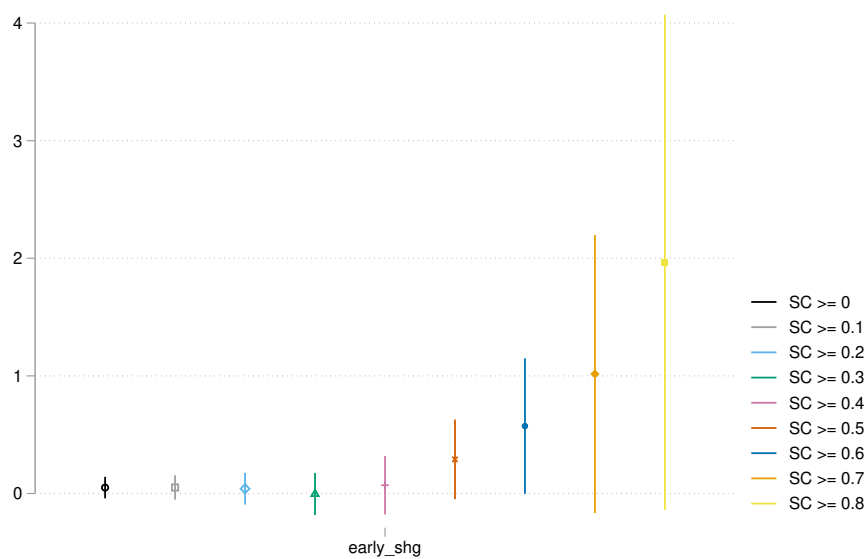
Notes: The outcome variable is the number of SC female candidates in Col 1. Cols 2–4 show the standardized years of education, the standardized asset based poverty score for candidates and the standardized log of Jati population. The poverty score is calculated by matching candidates to the SECC data and based on principal components analysis of households assets, which is subsequently aggregated to the GP level. The non-standardized means for the outcome variables is shown in the last row of the table. All regressions include village controls including GP total and SC population, area, and distance to the nearest town controls, as well as if the GP was reserved in the past for women. Robust standard errors are shown in brackets.

Table A9: Non-SC candidates running in treatment GPs and SC candidates running in treatment Wards

	(1)	(2)	(3)	(4)	(5)
	Total Cand	All Non-SC	Non-SC Women	All SC Ward	SC Ward Women
Treated	0.402 (0.709)	-1.719 (1.048)	-1.546 (1.067)	-0.002 (0.061)	0.046 (0.045)
Observations	179	179	179	1435	1435
Fixed Effects	Strata	Strata	Strata	Block	Block
GP Controls	Yes	Yes	Yes	Yes	Yes

Notes: The outcomes are Total Candidates, all other Non-SC candidates, and all other Non-SC Women candidates in Cols 1, 2, and 3 respectively. *Treated* is an indicator variable for GPs where SHGs were randomly introduced earlier in 2012. Cols 4 and 5 are regressions run at the Ward level where the SC population is atleast 1. The outcomes are all SC candidates and all SC Women candidates running in the Ward elections. All regressions include GP controls and Strata or Block fixed effects. Cols 1–3 include robust standard errors, and errors are clustered at the GP level in Cols 4 and 5.

Figure A5: SC Women participation in Ward Elections (by baseline SC share)



Notes: The outcome variable is the number of SC Women running in Wards. The figure shows the coefficient on earlier introduction of SHGs in samples restricted to wards with varying SC population shares. All regressions control for baseline village characteristics, and include block fixed effects and errors clustered at the GP level.

Table A10: Characteristics of candidates in the 2016 elections that did not match and those that matched to the SECC data

Variable	(1) Did not match		(2) Matched		T-test Difference (1)-(2)
	N	Mean/SE	N	Mean/SE	
OBC	646	0.571 (0.019)	1060	0.543 (0.015)	0.028
Other	646	0.022 (0.006)	1060	0.002 (0.001)	0.020***
SC	646	0.232 (0.017)	1060	0.283 (0.014)	-0.051**
ST	646	0.175 (0.015)	1060	0.172 (0.012)	0.003
Years of Education	640	6.183 (0.221)	1057	6.650 (0.171)	-0.467*
Age	645	39.932 (0.454)	1058	41.457 (0.351)	-1.525***
F-test of joint significance (F-stat)					6.727***
F-test, number of observations					1694

Notes: The value displayed for t-tests are the differences in the means across the groups. The value displayed for F-tests are the F-statistics. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A11: Candidate Characteristics: Experimental Estimates

	(1) Electability index	(2) Education	(3) Asset score	(4) Sub-caste size (log)
Panel A: Treatment effect				
Treated	0.140 (0.143)	0.072 (0.152)	0.055 (0.142)	-0.054 (0.138)
Observations	175	179	175	179
Panel B: By SC reservation				
Treated	0.295* (0.161)	0.188 (0.171)	0.227 (0.156)	-0.001 (0.170)
SC Reserved	-0.340 (0.216)	-0.347 (0.236)	-0.537 (0.331)	-0.296 (0.238)
Treated X SC Reserved	-0.512* (0.302)	-0.329 (0.323)	-0.497 (0.393)	-0.089 (0.328)
Observations	175	179	175	179

Notes: Each outcome is the GP-level average characteristic of the candidates matched to the SECC, in control-group standard deviations: the electability index (education, asset score, and log sub-caste size combined with weights estimated from a regression of an indicator for winning on the standardized components among control-GP candidates, with GP fixed effects), years of education, SECC asset score, and log sub-caste size (proxied by surname prevalence). *Treated* is an indicator for early-treatment GPs. Panel A reports the treatment effect; Panel B adds the GP's 2016 SC-reservation status and its interaction with treatment. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and block fixed effects, with robust standard errors. The index and asset score are available for the 175 GPs with at least one asset-matched candidate. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A12: Candidate Characteristics: Groups-Scaled and RD-RCT Estimates

	(1) Electability index	(2) Education	(3) Asset score	(4) Sub-caste size (log)
Panel A: Groups (OLS)				
Groups/100	0.057 (0.145)	0.220 (0.148)	-0.171 (0.139)	-0.011 (0.160)
SC Reserved	-0.175 (0.281)	0.085 (0.286)	-0.438 (0.400)	0.035 (0.289)
Groups/100 X SC Reserved	-0.450* (0.259)	-0.642*** (0.244)	-0.385 (0.345)	-0.417 (0.260)
Observations	175	179	175	179
Panel B: Groups (IV)				
Groups/100	0.677* (0.393)	0.447 (0.411)	0.520 (0.379)	-0.002 (0.403)
SC Reserved	0.384 (0.554)	0.138 (0.576)	0.113 (0.724)	-0.217 (0.609)
Groups/100 X SC Reserved	-1.052* (0.583)	-0.699 (0.595)	-0.976 (0.701)	-0.144 (0.631)
Observations	175	179	175	179
Panel C: RD-RCT				
Treated	0.489** (0.213)	0.273 (0.229)	0.486** (0.212)	-0.061 (0.221)
SC Reserved	0.332 (1.178)	0.111 (1.197)	0.643 (1.387)	-0.101 (1.157)
Treated X SC Reserved	-1.014 (1.131)	0.073 (1.396)	-1.436 (1.246)	0.726 (1.126)
Observations	175	179	175	179

Notes: Outcomes and controls as in Table A11. Panel A replaces the treatment indicator with the number of SHG groups formed by 2015 (in hundreds), interacted with 2016 SC-reservation status. Panel B instruments the number of groups and its interaction with treatment assignment and its interaction with reservation status, with small-sample standard errors. Panel C instruments reservation status and its interaction with treatment using whether the GP's running variable lies above its block's SC-population threshold, as in Eqs. (5) and (6), including the running variable as a control and with small-sample standard errors. Block fixed effects and robust standard errors throughout. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A13: Match Rates of SHG Members and Leaders with Female Candidates (Full Sample)

Category	Count/ Match Rate	Formula
Total SHG Members	2,758,505	
SHG Members Matched to Candidates	2,359	
Total SHG Leaders	30,515	
SHG Leaders Matched to Candidates	199	
Total Candidates	41,745	
Match Rate within SHG Members	0.08%	$\frac{\text{SHG Members Matched to Candidates}}{\text{Total SHG Members}}$
Match Rate within SHG Leaders	0.65%	$\frac{\text{SHG Leaders Matched to Candidates}}{\text{Total SHG Leaders}}$
Match Rate among Candidates (SHG Members)	5.65%	$\frac{\text{SHG Members Matched to Candidates}}{\text{Total Candidates}}$
Match Rate among Candidates (SHG Leaders)	0.48%	$\frac{\text{SHG Leaders Matched to Candidates}}{\text{Total Candidates}}$

Table A14: Match Rates of SHG Members and Leaders with Female Candidates (RCT Sample)

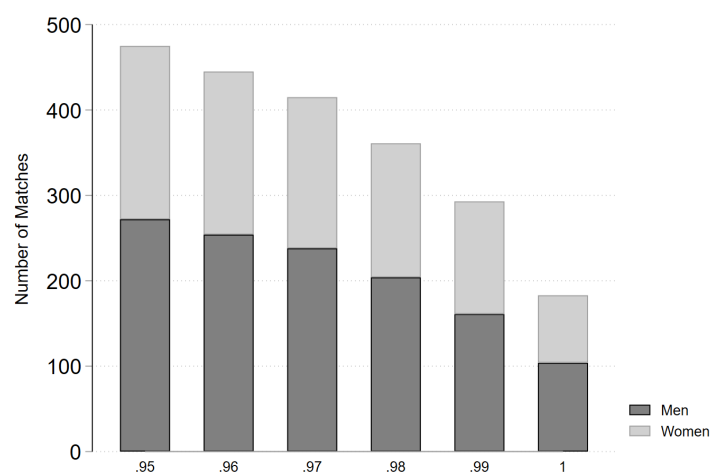
Category	Count/ Match Rate in Treatment	Count/ Match Rate in Control	Formula
Total SHG Members	71,574	54,492	
SHG Members Matched to Candidates	51	34	
Total SHG Leaders	894	294	
SHG Leaders Matched to Candidates	8	0	
Total Candidates	452	539	
Match Rate within SHG Members	0.071%	0.062%	$\frac{\text{SHG Members Matched to Candidates}}{\text{Total SHG Members}}$
Match Rate within SHG Leaders	0.89%	0.0%	$\frac{\text{SHG Leaders Matched to Candidates}}{\text{Total SHG Leaders}}$
Match Rate among Candidates (SHG Members)	11.3%	6.3%	$\frac{\text{SHG Members Matched to Candidates}}{\text{Total Candidates}}$
Match Rate among Candidates (SHG Leaders)	1.8%	0.0%	$\frac{\text{SHG Leaders Matched to Candidates}}{\text{Total Candidates}}$

Table A15: Benchmarking SHG Match Rates Against SHG Prevalence

	(1)	(2)	(3)
Panel A: SHG members			
	Match rate	Membership rate	Ratio (1)/(2)
Treated	0.085* (0.044)	0.096*** (0.015)	0.303 (0.185)
Observations	137	137	132
Control Mean	0.067	0.164	0.363
Panel B: SHG leaders			
	Leader match rate	Leader prevalence	Ratio (1)/(2)
Treated	0.0357 (0.0219)	0.0018*** (0.0002)	13.257* (6.751)
Observations	137	137	97
Control Mean	0.0000	0.0007	0.000

Notes: GP-level regressions on the 137 RCT GPs with at least one female candidate; *Treated* is an indicator for early-treatment GPs. In Panel A, Col 1 is the share of female candidates matched to an SHG member (match score above 0.98, group formed by 2015), Col 2 is the number of SHG members by 2015 divided by the GP's women (Census 2011 population scaled by the rural-Bihar sex ratio of 921 women per 1,000 men), and Col 3 is their ratio. Panel B repeats this for SHG leaders (VO presidents and secretaries of VOs formed by 2015, two per VO, divided by the same count of women). Under a random-draw null the match rate rises in proportion to prevalence, and the ratios in Col 3 do not move with treatment; Col 3 is undefined in GPs with no members (Panel A) or no leaders (Panel B) by 2015, which reduces its sample to 132 and 97 GPs. The levels of the ratios sit below one because the conservative name-match detects only a fraction of true members; the comparison across arms assumes equal detection in both, which holds by construction since the same algorithm and thresholds apply to both. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and block fixed effects, with robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Figure A6: Total Matches by Threshold (In Treatment GPs)



Notes: The figure shows the number of matches for females and males between candidate households in the SECC data and SHG member data. The X-axis shows the matching score thresholds above which we consider the names to be a match. The process for a successful match is as follows: we first restrict SECC data to candidate households, then both female member from SHG and father/husband name needs to match to a household in the SECC, and finally the SHG formation date for female member match needs to be in 2015 and earlier.

Table A16: Votes Polled, Total and for SC Candidates

	(1)	(2)	(3)	(4)	(5)
	ITT	IV	Het	RD-RCT	Votes per candidate
Panel A: Total votes polled					
Treated	-146.9 (124.4)		-195.7 (142.0)		-19.3 (37.9)
# of groups (instrumented)		-3.2 (2.9)			
SC Reserved			-428.4*** (164.1)		-197.5*** (71.6)
Treated X SC Reserved			413.9* (236.1)		46.2 (91.4)
Treated X SC Reserved (RD-RCT)				686.5 (1506.5)	
Observations	176	176	176	176	176
Control Mean	4451.5	4451.5	4451.5	4451.5	462.2
Panel B: Votes polled for SC candidates					
Treated	526.1* (274.1)		-48.4 (61.1)		-42.1 (49.5)
# of groups (instrumented)		11.5* (6.2)			
SC Reserved			3922.7*** (168.9)		124.9*** (43.8)
Treated X SC Reserved			397.4* (219.7)		60.1 (73.0)
Treated X SC Reserved (RD-RCT)				805.2 (519.2)	
Observations	176	176	176	176	77
Control Mean	721.1	721.1	721.1	721.1	228.0

Notes: The sample is the 176 RCT GPs with a recorded poll in the 2016 GP-head election; *Treated* is an indicator for early-treatment GPs. The outcome in Panel A is total votes polled. In Panel B it is votes polled for SC candidates, summed from candidate-level records: in SC-reserved GPs every candidate is de jure SC, so the outcome there coincides with total votes, while in unreserved GPs it isolates the votes cast for SC candidates. Col 1 is the experimental estimate. Col 2 instruments the number of SHG groups formed by 2015 with treatment assignment (standard errors with a small-sample correction). Col 3 adds the GP's 2016 SC-reservation status and its interaction with treatment. Col 4 instruments reservation status and its interaction with treatment using whether the GP's running variable lies above its block's SC-population threshold, as in Eqs. (5) and (6). Col 5 repeats the Col 3 specification with votes per candidate as the outcome, built from candidates with recorded votes; 77 GPs have at least one SC candidate with recorded votes, which sets the Panel B sample in that column. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and block fixed effects, with robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A17: Winner Identity and SHG Connection

	(1) Treatment effect	(2) Control mean	(3) Obs.
Panel A: Winner identity, margin-unreserved seats			
Woman not women-reserved	0.065 (0.082)	0.170	101
SC not SC-reserved	no SC winner in any of the 144 seats, either arm		
SC woman not SC-and-women-reserved	0.028 (0.030)	0.035	165
Panel B: Winner's SHG connection			
From SHG member household	0.011 (0.061)	0.184	168
From SHG leader household	-0.001 (0.018)	0.011	168
Is herself an SHG member	0.046 (0.053)	0.115	168

Notes: Treatment effects on the identity of the 2016 GP-head winner and on the winner's SHG connection, in probability units; *Treated* is an indicator for early-treatment GPs. Panel A restricts to seats where the identity is not guaranteed by a quota: winner is a woman among GPs not reserved for women; winner is SC among caste-unreserved GPs (no regression is estimable – no SC candidate wins any of the 144 caste-unreserved seats for which we observe a winner, in either arm); winner is an SC woman among GPs not reserved for both SCs and women. Panel A regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and block fixed effects. Panel B uses the SECC-SHG match at the GP level: whether the winner comes from an SHG member household, from an SHG leader household (VO office-holders), or is herself a matched SHG member (match score ≥ 0.99 , groups formed by 2015; Appendix IX); these regressions include block fixed effects. Robust standard errors. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A18: Winner Characteristics

	(1)	(2)	(3)	(4)
	Age	Education	Asset score	Sub-caste size (log)
Panel A: Treatment effect				
Treated	-0.065 (0.181)	0.113 (0.172)	0.505 (0.362)	-0.002 (0.161)
Observations	176	174	88	139
Panel B: By SC reservation				
Treated	0.046 (0.168)	0.089 (0.176)	0.234 (0.314)	0.005 (0.175)
SC Reserved	-0.270 (0.333)	-0.409 (0.319)	-0.397 (0.358)	-0.009 (0.311)
Treated X SC Reserved	-0.208 (0.435)	0.319 (0.425)	-0.202 (0.682)	0.093 (0.394)
Observations	176	174	88	139

Notes: Outcomes are the 2016 GP-head winner's age, years of education, SECC asset score, and log sub-caste size (surname prevalence), standardized against control-group winners. *Treated* is an indicator for early-treatment GPs. Panel A reports treatment effects with randomization-strata fixed effects; Panel B adds 2016 SC-reservation status and its interaction with treatment, with block fixed effects. All regressions include the GP controls of Table A17 and robust standard errors. Characteristics are observed for 176 (age), 174 (education), 139 (sub-caste), and 88 (asset score; SECC-matched winners) GPs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A19: Effective Number of candidates and Margin of Victory in treatment GPs

	(1)	(2)
	ENC	MoV
Treated	0.038 (0.372)	0.009 (0.013)
Observations	176	176
Fixed Effects	Strata	Strata
Control Mean	6.28	0.08

Notes: The outcome variables are the effective number of candidates (ENC) and margin of victory (MoV) between winner and runner-up candidate in the local village elections in 2016. All columns show the results for the RCT, where Treated are GPs where the SHG program was introduced earlier. All regressions include strata fixed effects, village controls including GP total and SC population, area, and distance to the nearest town controls, as well as if the GP was reserved in the past for women. Robust standard errors are shown in brackets.

Table A20: Number of SC and SC Women candidates running in treatment GPs in the 2021 local elections

All SCs					
	(1)	(2)	(3)	(4)	(5)
	RCT	IV	RCT Unreserved	RCT Het	RD-RCT
Treated	0.772 (0.559)		-0.135 (0.241)		0.099 (0.216)
num_groups_2020		0.141 (0.163)			
1.Treatment				0.032 (0.182)	
SC_Reserved				7.512*** (0.848)	
Treated X SC Reserved				-1.075 (1.076)	
Treated X SC Reserved					-1.001 (1.325)
Observations	167	167	59	167	167
Fixed Effects	Strata	Strata	None	Block	Block
Sample	All	All	All Unreserved	All	All
Control Mean	1.48	1.48	0.54	1.48	1.48
SC Women					
	(1)	(2)	(3)	(4)	(5)
	RCT	IV	RCT Unreserved	RCT Het	RD-RCT
Treated	0.722* (0.377)		0.045 (0.209)		-0.069 (0.185)
num_groups_2020		0.132 (0.137)			
1.Treatment				0.073 (0.170)	
SC_Reserved				2.564** (1.042)	
Treated X SC Reserved				1.236 (1.461)	
Treated X SC Reserved					1.145 (1.034)
Observations	167	167	108	167	167
Fixed Effects	Strata	Strata	Strata	Block	Block
Sample	All	All	Caste Unreserved	All	All
Control Mean	0.55	0.55	0.19	0.55	0.55

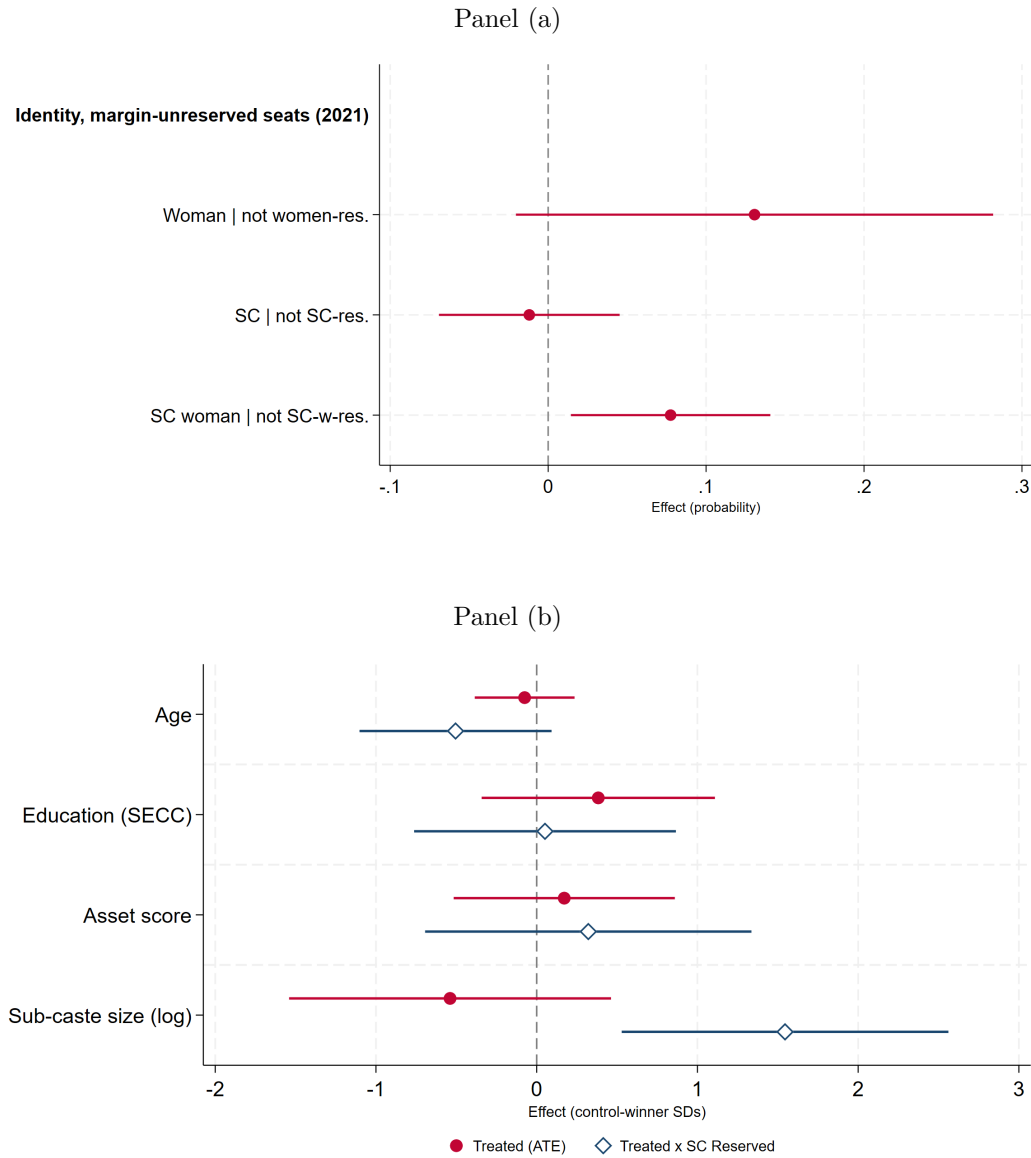
Notes: The outcome variable in the top panel is the number of SC candidates running in the local village elections in 2021. The bottom panel shows the same regressions with SC women as the outcome variable. Col 1 shows the results for the RCT. Col 2 shows the number of groups in 2020 instrumented using the *Treated* instrument. Col 3 shows results for the sample excluding GPs reserved for SCs, OBCs and Women in 2016. Col 4 shows results using the SC reservation indicator in 2016 and the interaction with earlier SHG rollout. Col 5 shows the results from the RD-RCT specification i.e. shows the coefficient on the interaction term above the SC reservation cutoff. Fixed Effect denotes whether the paired RCT strata or block fixed effects were used. All regressions include village controls including GP total and SC population, area, and distance to the nearest town controls, as well as if the GP was reserved in the past for women. Robust standard errors are shown in brackets.

Table A21: SC-Woman Winners in 2021, by 2016 Seat Reservation

	(1)	(2)	(3)
	Control	Treated	Fisher exact p
SC and women reserved	3/3	8/8	–
SC reserved only	1/10	6/10	0.057
Women reserved only	1/38	0/24	1.000
Unreserved	0/34	0/40	–
All GPs	5/85	14/82	0.028

Notes: Each cell reports the number of GPs whose 2021 GP-head winner is an SC woman over the number of GPs with winner data, by treatment arm and the GP's 2016 seat reservation, which was frozen through the 2021 election. In GPs reserved for SC women, the winner is an SC woman by construction. Col 3 reports two-sided Fisher exact p -values for the treatment–control difference within each row; dashes mark rows with no variation in the outcome.

Figure A7: Who Wins in 2021



Notes: Outcomes describe the winner of the 2021 GP-head election (recorded in the 2022 electoral data), observed for 167 of the 179 RCT GPs; *Treated* is an indicator for early-treatment GPs. Reservation status is frozen at its 2016 assignment, which binds: every one of the 11 GPs reserved for both SCs and women elected an SC woman in 2021. Panel (a) plots treatment effects, in probability units, on winner identity in seats where the identity is not guaranteed by a quota; unlike in 2016, when no SC candidate won any caste-unreserved seat, five SC candidates won open seats in 2021, so the SC margin is estimable. The SHG-connection outcomes of the 2016 figure are unavailable, as 2021 candidates are not name-matched to SHG rolls. Panel (b) plots effects on the winner's age (from the electoral records), years of schooling and asset score (both from the 2011–12 SECC, for the 81 GPs whose 2021 winner is matched to an SECC household), and log sub-caste size (households of the winner's jati in the GP), standardized against control-group winners; filled circles are treatment effects with randomization-strata fixed effects, hollow diamonds the interaction of treatment with 2016 SC-reservation status, with block fixed effects. All regressions include GP controls (2006 SC population, 2011 Census area and population, distance to the nearest town, and 2006 female-reservation status) and robust standard errors; bars are 90% confidence intervals. Full regression results appear in Tables A22 and A23.

Table A22: Winner Identity in 2021

	(1) Treatment effect	(2) Control mean	(3) Obs.
Winner identity, 2021, margin-unreserved seats			
Woman not women-reserved	0.131 (0.092)	0.114	94
SC not SC-reserved	-0.012 (0.035)	0.042	136
SC woman not SC-and-women-reserved	0.078** (0.038)	0.024	156

Notes: Specifications as in Panel (a) of Figure A7: linear probability models with GP controls and block fixed effects, robust standard errors, reservation frozen at 2016. The SC-woman result is concentrated where it can occur: among the 20 SC-reserved GPs that carry no women's quota, 6 of 10 treated GPs elected an SC woman against 1 of 10 in control (Fisher exact $p = 0.057$, two-sided); no SC woman won an open seat in either arm. In 2016 the same margin was 2.8 percentage points and insignificant. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A23: Winner Characteristics in 2021

	(1)	(2)	(3)	(4)
	Age	Education (SECC)	Asset score	Sub-caste size (log)
Panel A: Treatment effect				
Treated	-0.075 (0.189)	0.383 (0.441)	0.171 (0.418)	-0.539 (0.609)
Observations	167	81	81	78
Panel B: By SC reservation				
Treated	0.016 (0.187)	0.283 (0.272)	0.258 (0.351)	-0.630* (0.327)
SC Reserved	-0.089 (0.217)	-0.632** (0.292)	-0.637* (0.354)	-1.195*** (0.360)
Treated X SC Reserved	-0.505 (0.363)	0.052 (0.495)	0.321 (0.617)	1.545** (0.618)
Observations	167	81	81	78

Notes: Outcomes are the 2021 winner's age (from the electoral records, $N = 167$ GPs), years of schooling and asset score (from the 2011–12 SECC, via the match of 2021 candidates to SECC households; $N = 81$ GPs with a matched winner), and log sub-caste size (households of the winner's jati in the GP), standardized against control-group winners. Affidavit-based education is unavailable for the 2021 election. Panel A reports treatment effects with randomization-strata fixed effects; Panel B adds 2016 SC-reservation status and its interaction with treatment, with block fixed effects. All regressions include the GP controls of Figure A7 and robust standard errors. Estimates using SECC characteristics are similar in the 70 GPs where the matched winner's gender agrees with the electoral records. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A24: Downstream outcomes in treatment GPs: Public Goods and NREGA

	(1)	(2)	(3)	(4)	(5)
	Total Roads	Total schools	School Quality	Tap water	NREGA SC workdays
Treated	-0.529 (0.435)	0.197 (0.424)	34.329 (59.343)	0.097 (0.066)	-1343.014 (1617.818)
SC_Reserved	-0.057 (0.779)	-1.029 (0.888)	-2.358 (25.922)	0.034 (0.127)	435.253 (2971.783)
Treated X SC Reserved	-0.134 (1.049)	1.143 (1.056)	-83.129 (98.900)	-0.125 (0.155)	-882.047 (3753.250)
Observations	179	179	178	178	179
Fixed Effects	Strata	Strata	Strata	Strata	Strata
GP Controls	Yes	Yes	Yes	Yes	Yes

Notes: The table shows differences in the match rates for a matching score of 0.99. Col 1. shows that father/husband of SHG members and leaders between SECC and SHG treatment and control GPs.

Table A25: Downstream outcomes in treatment SHGs: Access to Loans

	(1)	(2)	(3)
	ICF	HRF	FSF
Treated	0.101*** (0.024)	0.247*** (0.035)	0.184*** (0.030)
SC_Reserved	-0.059 (0.055)	0.030 (0.070)	-0.068 (0.043)
Treated X SC Reserved	0.012 (0.067)	0.005 (0.090)	0.011 (0.062)
Observations	175	175	175
Fixed Effects	Block	Block	Block
GP Controls	Yes	Yes	Yes
Control Mean	0.76	0.36	0.56

Notes: Cols 1-3 shows the share of SHGs that have received 3 different kinds of loans by 2024. In Col 1, the outcome is the share that have received the Initial Capitalization Fund. In Col 2 the outcome is the Health Risk Fund. In Col 3, the outcome is the share that have received the Food Security Fund. All columns show results using the SC reservation indicator (*SC_Reserved*) in 2016 and the interaction with earlier SHG rollout (*Treated*). All regressions include village controls including GP population, area, and distance to the nearest town controls, block fixed effects, and robust standard errors shown in brackets. All columns show the control group mean at the bottom.

Table A26: Characteristics of candidates in the 2016 elections that did not match and those that matched to SHG members

Variable	N	(1) 0 Mean/(SE)	N	(2) 1 Mean/(SE)	N	(1)-(2) Pairwise t-test Mean difference
OBC	39185	0.523 (0.003)	2560	0.387 (0.010)	41745	0.136***
Other	39185	0.243 (0.002)	2560	0.075 (0.005)	41745	0.168***
SC	39185	0.224 (0.002)	2560	0.530 (0.010)	41745	-0.306***
ST	39185	0.011 (0.001)	2560	0.009 (0.002)	41745	0.002
Years of Education	38965	4.754 (0.025)	2539	4.764 (0.098)	41504	-0.010
Age	39172	39.148 (0.055)	2559	39.424 (0.214)	41731	-0.276
F-test of joint significance (F-stat)						263.970***
F-test, number of observations						41490

Significance: ***=.01, **=.05, *=.1.

Table A27: 2SLS Regressions using Years of Exposure

	(1)	(2)	(3)	(4)	(5)	(6)
	Total	SC	Women	SC Women	Non-SC	Male
Years of exposure	0.216 (0.327)	1.021** (0.497)	-0.163 (0.478)	0.580* (0.301)	-0.805 (0.496)	0.379 (0.493)
Observations	179	179	179	179	179	179
Fixed Effects	Strata	Strata	Strata	Strata	Strata	Strata
Control Mean	12.14	2.74	6.22	0.97	9.40	5.92

Notes: The outcomes are the total, SC, Women, SC Women, Non-SC and Male number of candidates that ran in the local elections in 2016. All columns show the years of exposure to the SHG program in 2015 instrumented using the earlier SHG program rollout indicator (Treated). All regressions have strata fixed effects and robust standard errors in brackets. Standard errors include the small-sample degrees-of-freedom correction for the absorbed strata dummies.

Table A28: Balance on GPs with different levels of exposure (Census 2001)

Variable	(1) Below 3 yrs	(2) Above 3 Yrs	(3) Difference
GP population (2001 Census)	8,975.093 (3,231.849)	9,384.815 (3,564.613)	592.094*** (143.706)
SC population share (2001 Census)	0.142 (0.062)	0.158 (0.070)	0.002 (0.002)
Number of villages in GP (2001 Census)	5.468 (4.005)	5.070 (4.778)	0.063 (0.144)
GP total area (sq km), 2001	1,072.112 (866.132)	1,058.793 (734.571)	20.047 (30.001)
Distance to District Headquarters (km), 2001	32.235 (18.331)	30.894 (18.394)	-0.600** (0.296)
Distance to Nearest Statutory Town (km), 2001	19.897 (12.859)	21.140 (13.892)	-0.708*** (0.272)
Total Educational Facilities (Census 2001)	3.563 (1.903)	3.330 (1.985)	-0.033 (0.073)
Primary Health Sub Centres (Census 2001)	0.397 (0.655)	0.464 (0.679)	0.033 (0.027)
Post Office (Census 2001)	1.015 (0.808)	1.025 (0.777)	-0.025 (0.035)
Bank Facilities (Census 2001)	0.329 (0.590)	0.333 (0.620)	0.025 (0.027)
Power Supply (Census 2001)	1.713 (1.895)	1.797 (1.890)	0.025 (0.069)
Paved Road (Census 2001)	1.865 (1.744)	1.771 (1.618)	-0.010 (0.069)
Mud Road (Census 2001)	4.080 (2.817)	3.854 (3.424)	-0.030 (0.105)
Child Welfare Centre (Census 2001)	0.057 (0.398)	0.064 (0.527)	0.021 (0.018)
Bus facilities (Census 2001)	0.553 (0.954)	0.534 (0.917)	0.016 (0.039)
Mean Village Income (Census 2001)	607592.875 (9.473e+06)	274025.156 (3.014e+06)	-75719.406 (347856.594)
Mean Village Expenditure (Census 2001)	613333.750 (1.348e+07)	206243.703 (2.302e+06)	-4.477e+05 (502857.719)
Middle School Facility (Census 2001)	1.160 (1.024)	1.211 (1.032)	0.033 (0.044)
Medical Facility (Census 2001)	0.898 (1.102)	0.881 (0.968)	0.008 (0.042)
Tap Water (Census 2001)	0.149 (0.565)	0.131 (0.605)	0.015 (0.024)
Well (Census 2001)	3.871 (3.038)	3.608 (3.768)	0.019 (0.103)
Tank (Census 2001)	0.558 (1.305)	0.450 (1.368)	-0.066 (0.049)
Phone Lines (Census 2001)	0.548 (0.841)	0.578 (0.829)	0.072** (0.034)
Observations	3,675	1,635	5,310

Table shows balance between GPs exposed for under 3 years with GPs exposed for greater than 3 years within the RD bandwidth on census 2001 variables.

Table A29: Balance on GPs with different levels of exposure (Census 2001)

Variable	(1) No exposure	(2) Below 3 Yrs	(3) Difference
GP population (2001 Census)	10,945.496 (6,172.084)	8,975.093 (3,231.849)	-1,554.358*** (141.418)
SC population share (2001 Census)	0.131 (0.069)	0.142 (0.062)	0.005*** (0.002)
Number of villages in GP (2001 Census)	4.406 (3.959)	5.468 (4.005)	0.732*** (0.111)
GP total area (sq km), 2001	1,319.036 (1,283.827)	1,072.112 (866.132)	-266.500*** (29.674)
Distance to District Headquarters (km), 2001	32.079 (17.603)	32.235 (18.331)	-0.927*** (0.245)
Distance to Nearest Statutory Town (km), 2001	21.957 (14.247)	19.897 (12.859)	-1.440*** (0.225)
Total Educational Facilities (Census 2001)	2.923 (1.901)	3.563 (1.903)	0.425*** (0.058)
Primary Health Sub Centres (Census 2001)	0.451 (0.771)	0.397 (0.655)	-0.065*** (0.023)
Post Office (Census 2001)	1.059 (0.852)	1.015 (0.808)	-0.055* (0.028)
Bank Facilities (Census 2001)	0.356 (0.630)	0.329 (0.590)	-0.021 (0.022)
Power Supply (Census 2001)	1.326 (1.644)	1.713 (1.895)	0.422*** (0.053)
Paved Road (Census 2001)	1.355 (1.401)	1.865 (1.744)	0.475*** (0.054)
Mud Road (Census 2001)	3.522 (2.928)	4.080 (2.817)	0.507*** (0.081)
Child Welfare Centre (Census 2001)	0.063 (0.597)	0.057 (0.398)	-0.020 (0.015)
Bus facilities (Census 2001)	0.446 (0.759)	0.553 (0.954)	0.082*** (0.030)
Mean Village Income (Census 2001)	121485.547 (1.581e+06)	607592.875 (9.473e+06)	429525.750 (284954.250)
Mean Village Expenditure (Census 2001)	107648.625 (1.523e+06)	613333.750 (1.348e+07)	290003.781 (405729.594)
Middle School Facility (Census 2001)	1.304 (1.213)	1.160 (1.024)	-0.169*** (0.036)
Medical Facility (Census 2001)	0.792 (0.927)	0.898 (1.102)	0.044 (0.033)
Tap Water (Census 2001)	0.138 (0.479)	0.149 (0.565)	-0.034* (0.018)
Well (Census 2001)	3.227 (3.175)	3.871 (3.038)	0.411*** (0.080)
Tank (Census 2001)	0.524 (1.280)	0.558 (1.305)	0.044 (0.039)
Phone Lines (Census 2001)	0.479 (0.712)	0.548 (0.841)	0.076*** (0.026)
Observations	1,472	3,675	5,147

Table shows balance between GPs never exposed to SHGs and those exposed for under 3 years within the RD bandwidth on census 2001 variables.

VIII. Power and Minimum Detectable Effects

To assess the sensitivity of our main experimental design, we compute minimum detectable effects (MDEs) using the residual standard deviations from regressions that absorb the original pairwise randomization strata. This ensures that our power calculations correspond exactly to the specification used in the paper. Using the 179 GPs in the experiment (89 treated, 90 control), the residual variance after absorbing strata fixed effects is modest for all three key outcomes: the residual standard deviation is 4.88 for the number of Scheduled Caste (SC) candidates, 4.42 for the number of women candidates, and 2.79 for the number of SC women candidates. Plugging these into standard two-sample power calculations (with a significance level of 0.05, two-sided, and power of 0.80), we find that the design can detect differences of 2.06 SC candidates, 1.86 women candidates, and 1.17 SC women candidates per GP. Given typical magnitudes in the literature on political entry (Gulzar, Haas, and Pasquale 2023, Pande 2003), these detectable effects are quite reasonable. We therefore conclude that the experiment is well powered to detect meaningful treatment effects for our main outcomes of interest.

IX. Matching description

A. Matching SECC to candidates

Matching Methodology We matched data on all candidates (N=2108) in the RCT sample of 179 GPs with data on all citizens using the 2012 Socioeconomic Caste Census (SECC). Candidates were string matched to citizens based on characteristics, including name, guardian’s name, age, caste, and sub-caste (proxied by last name).

Match Rates We were able to match with high confidence (i.e., matching scores above 0.98) 1287 out of 2108 candidates, yielding a match rate of 61%.

B. Matching SHG member details to candidates

Matching Methodology We obtained SHG membership data for 5,393 of the 6,124 GPs where at least one woman contested the 2016 local elections. This restricted our candidate sample to 41,745 female candidates out of the total of 47,101 female candidates. We matched on name, father’s or husband’s name, and caste category to obtain a matching score. We used a high threshold of 0.98 on the matching score to determine successful matches.

Match Rates Overall, we match 2,359 of the 2,758,505 SHG members (excluding leaders) from 2015 to candidate lists, yielding a match rate of 0.08%. For higher-level SHG leaders (Presidents and Secretaries of VOs and CLFs), we match 199 of 30,515 individuals, corresponding to a match rate of 0.65%.

In the RCT sample, 991 female candidates contested elections in 137 of the 179 GPs.¹⁹ The match rate in treated GPs for SHG members is 0.071% and for SHG leaders is 0.89% – that is, the share of members (leaders) who appear on the candidate lists.

Tables Showing Match Rates See Table A13 for match rates in the full sample, and Table A14 for match rates in the RCT sample. Table A26 shows that candidates who matched with SHG members, and those who did not, are similar in demographics such as age and education and only differ on caste (since the SHG program targeted SC households).

19. This includes 63 treatment and 74 control GPs.

X. A Simple Framework of Political Entry

We develop a simple framework of political entry to interpret how SHGs and reservation shape the decisions of SC households to contest GP-head and ward elections. Following Dal Bó and Finan (2018), we treat a candidate’s valence—a bundle of traits valued by all voters—as the key supply-side characteristic.

We model entry as a household decision rather than an individual one. Political attitudes and participation are shaped within families, with household support central to political ambition (Verba, Nie, and Kim 1978; Beck and Jennings 1982; Jennings and Niemi 1981; Fox and Lawless 2010). In South Asia, men frequently make candidacy decisions for the household, and brokers target families, not individuals (Prillaman 2023b); running for office is costly, and households often designate one member for the role (Burns, Schlozman, and Verba 2001); political dynasties underscore the household as the unit of political investment (Cruz, Labonne, and Querubin 2017; Verma 2021; George 2024; Bamezai 2025). The assumption is particularly apt for SHGs: membership is restricted to women, but the benefits—savings, community ties, leadership positions—accrue to the household. Consistent with this, SHG-linked households field slightly more male than female candidates in both arms (Figure A6).

Set-up. Household h is described by its valence $v_h \in [0, \bar{v}]$: an index of electability, the bundle of household traits that raises a candidate’s chances at the polls. A seat is characterized by the value of holding the office, $B \geq 0$; the net cost of contesting it, $C > 0$ (campaign outlays and forgone earnings); and its reservation status θ , which determines eligibility. Standing for office also carries a small non-office benefit $b \geq 0$ —the status, visibility, and service value of candidacy itself, received whether or not the candidate wins. The probability of victory $P(v; \theta) \in [0, 1]$ is strictly increasing in v wherever it is interior. Household h fields an eligible member if and only if

$$P(v_h; \theta) B + b \geq C. \quad (7)$$

Given monotonicity, entry takes a threshold form: there exists v^* such that all households with $v_h \geq v^*$ contest. The distinction between B and b matters below: B is scaled by the chance of winning, b is not.

SHGs. SHG exposure benefits member households through three channels: *liquidity* (savings and credit ease C); *networks and visibility* (wider ties, often crossing sub-castes, raise P); and *demonstrated leadership* (positions in the SHG hierarchy publicly signal organizational ability, raising P). Write the SHG shift as $P(v; \theta) + \delta$ with $\delta > 0$, and a cost reduction $C - \Delta_C$ with $\Delta_C \geq 0$. Both relax (7), lowering the entry threshold: $v^*(\theta, \text{SHG}) < v^*(\theta, 0)$.

Remark 1. The visibility and leadership channels are naturally strongest for SHG officeholders, so δ is plausibly largest for leader households. We do not derive this within the framework; we examine it directly in Section E.

Reservation. Caste reservation restricts the field to SC households, discretely raising P for every SC household: $P(v; \theta = 1) > P(v; \theta = 0)$, so $v^*(1, \cdot) < v^*(0, \cdot)$. A gender quota operates differently: it restricts which member a household may field, but does not change the caste composition of the field an SC household faces. In a caste-unreserved seat, a gender quota therefore leaves an SC household’s winning odds essentially unchanged.

Which seats respond. The framework’s content comes from where the entry margin (7) can move. For the GP headship, B is large—the Mukhiya controls funds, works, and appointments—and C is substantial relative to b : canvassing a constituency of roughly 10,000 residents requires real outlays, so status alone does not drive head-seat candidacy ($b < C$). Entry at the head seat therefore turns on $P \cdot B$.

Assumption 1 (No SC viability in open head seats). *In caste-unreserved GP-head seats, $P(v; 0) B + b + \delta B < C$ for all $v \leq \bar{v}$ among SC households: even the highest-valence SC household, with the SHG boost, falls short of the entry margin.*²⁰

Assumption 1 is disciplined by the data rather than assumed for convenience: in control GPs no SC candidate wins a caste-unreserved head seat and fewer than 10% of candidates are SC, irrespective of women’s reservation; across both arms of the experiment, none of the 144 winners of caste-unreserved seats is SC. Under it, SHG exposure moves no SC household across the threshold in open seats—with or without a gender quota, since a gender quota does not alter the field’s caste composition. In SC-reserved seats, by contrast, P is interior and (7) bites: the SHG shift lowers v^* , and SC candidacy rises. With both quotas, the household’s fielded member must be an SC woman, so entry there takes the form of SC women’s candidacy. Marginal entrants—households with v between the new and old thresholds—have lower valence than every inframarginal contestant, so where entry rises, the average valence of the candidate pool falls.

Entry without victory.

Proposition 1 (Induced entrants rarely win). *Consider any policy change that relaxes (7) (an increase δ in P , a reduction Δ_C in C , or both), moving the threshold from v_0^* to $v_1^* < v_0^*$. Every induced entrant ($v \in [v_1^*, v_0^*)$) has a strictly lower probability of victory than every household that would have contested anyway, and that probability is bounded above by*

$$P(v_0^*; \theta) = \frac{C - b}{B}.$$

Proof. Inframarginal contestants have $v \geq v_0^* > v'$ for any induced entrant v' , and P is strictly increasing in v , giving the comparison. At the pre-change threshold, (7) holds

20. A minimal voting microfoundation delivers this corner. Suppose voting is caste-first: each voter supports a candidate from her own caste bloc when one stands, ranking within-bloc candidates by valence (Chandra 2004; Banerjee and Pande 2007). With SC households a minority of the electorate, an SC candidate in a caste-open seat then loses to any dominant-caste rival regardless of valence, so $P(v; 0) = 0$ for all v ; and because SHG networks and visibility deliver votes within the bloc, the boost δ cannot overturn the bloc arithmetic. The entry condition (7) reduces to $b \geq C$, which fails at the head seat, where campaign costs are substantial. Assumption 1 thus follows from minority demography and bloc voting alone.

with equality: $P(v_0^*; \theta)B + b = C$. Any induced entrant has $v < v_0^*$, hence $P(v; \theta) < P(v_0^*; \theta) = (C - b)/B$. $\square$

The same threshold logic that generates entry effects thus caps their electoral consequences. The households SHGs draw into politics enter precisely because the bar fell to meet them; they carry the lowest winning odds in the field, so the identity and characteristics of winners move at most second-order. Entry responds; victory does not. Section H tests this directly; its predictors-of-winning estimates are consistent with the mechanism: education and assets predict victory within GPs.

Ward elections. Ward constituencies are tiny—roughly 900 residents against 10,000 for a GP—and often caste-homogeneous. In 2016, ward members had virtually no authority: while every Panchayat elected 13–14 ward members alongside one GP head, power was concentrated entirely in the head’s office, and contemporary accounts describe ward members as excluded from information, resources, and decision-making (Sharan 2021).²¹ Financial powers and functions were devolved to ward-level bodies only after the 2016 elections (Bamezai et al. 2026).

Assumption 2 (Ward stakes in 2016). *For ward seats, the office value is negligible and contesting is nearly free: $B_W \approx 0$ and $C_W \approx 0$, with $b \geq C_W$ for some households in each ward.*

Proposition 2 (Ward entry: full seats, no response). *Under Assumption 2, ward contestants are of two types. (i) Households with $b \geq C_W$ enter regardless of P : seats fill even though the office is worth nothing, and this entry is invariant to SHG exposure or reservation, which operate through P and C . (ii) A household with $b < C_W$ enters only if $P \geq (C_W - b)/B_W$, a bar that rises to one as $B_W \rightarrow 0$. Hence for $B_W \approx 0$, SHG exposure does not change ward candidacy.*

Proof. For type (i), (7) holds at $P = 0$ and continues to hold under any increase in P ; the decision does not depend on P , hence not on δ ; and with $C_W \approx 0$ there is no cost margin for Δ_C to relax. For type (ii), (7) requires $P B_W \geq C_W - b > 0$; as $B_W \rightarrow 0$ the required P exceeds one, so no such household enters, with or without the SHG shift δ . $\square$

Filled ward seats are therefore evidence that the status value of candidacy exceeds its near-zero cost—they say nothing about the value of the office, and they are fully consistent with a null effect of SHGs on ward entry.

Corollary 1 (Where ward entry can respond). *If instead B_W is small but positive, type-(ii) entry is possible only where victory is near-certain, $P \rightarrow 1$: wards where SC households form a dominant share of the population. Any SHG effect on ward candidacy is confined to such wards.*

21. As one Dalit ward member put it, “power ha[d] not been given to ward members. This [is] why [the] Mukhiya acts in an arbitrary manner. Ward members should be given financial and other powers for development work . . . development can pick up if financial powers are given to ward members.” Another noted simply: “No Mukhiya . . . wants to share power with the other representatives in the Panchayati Raj system.” (Sachchidananda 2007)

Corollary 1 matches the one place the ward data move: SC women’s candidacy responds to SHG exposure only in wards with a very high SC population share (Appendix Figure A5).

Summary of predictions. The framework yields four predictions.

1. In caste-unreserved head seats, SHG exposure does not change SC candidacy—whether or not the seat carries a gender quota—and produces no SC winners (Assumption 1). *Tested in Sections C and H.*
2. In SC-reserved seats, SHG exposure raises SC candidacy; where a women’s quota compounds the caste quota, the entrants are SC women. *Tested in Section C.*
3. Where entry rises, the valence of the candidate pool falls: induced entrants have lower electability than every inframarginal contestant. *Tested in Section D.*
4. Induced entrants rarely win (Proposition 1): winner identity and characteristics are unchanged even where candidacy responds. At the ward level, entry does not respond at all despite filled seats (Proposition 2), except possibly where SC population shares make victory near-certain (Corollary 1). *Tested in Section H and Appendix Table A9.*

What the framework abstracts from. The framework takes P as a reduced-form function of own valence and the seat’s reservation status, abstracting from strategic interaction among entrants; free-entry congestion—each entrant lowering rivals’ winning odds—would reinforce, not overturn, the null predictions. The household is treated as unitary, and b as a primitive. We view the framework as an organizing device for the empirical results, not as a contribution in itself.

XI. Long-run exposure

The RCT documents that an additional two years of SHG activity can shift candidacy, with effects concentrated in reserved GPs. In per-year terms, instrumenting years of exposure with treatment assignment, each year of SHG exposure raises SC candidacy by 1.0 candidates and SC-women candidacy by 0.6 (Appendix Table A27). We next examine whether entry scales with longer exposure in the full cross-section of GPs in Bihar. For each GP, we observe the year of first SHG presence. We use this to define years of exposure to SHGs as of 2016. We group exposure into bins (≤ 3 years, > 3 years), treating GPs that had not received SHGs by 2016 as the omitted category. We do so because typically after 3 years of initial SHG entry, SHGs can be considered to be mature: by this time, SHGs have received additional loans and are federated under VOs.

We estimate a difference-in-discontinuities style specification within a symmetric window around the SC-reservation cutoff, absorbing block fixed effects and reporting robust

standard errors:

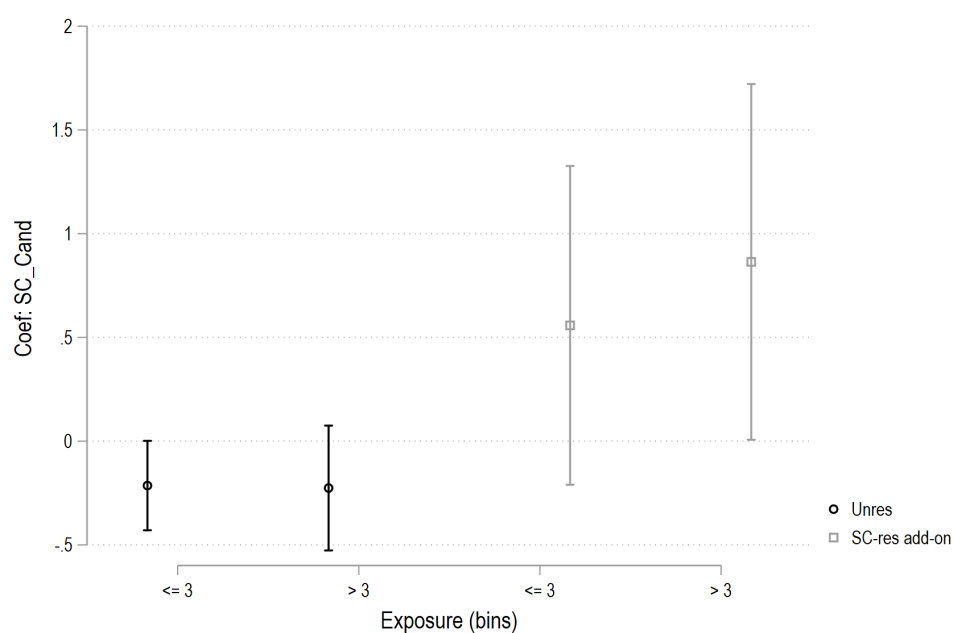
$$\begin{aligned}
SC_Cand_{gb} = & \alpha + \sum_{e \in \mathcal{E}} \beta_e \mathbb{I}\{\text{exp_bin} = e\}_g + \gamma SCRes_g + \sum_{e \in \mathcal{E}} \delta_e \mathbb{I}\{\text{exp_bin} = e\}_g \times SCRes_g \\
& + \rho Dist_g + \sum_{e \in \mathcal{E}} \kappa_e \mathbb{I}\{\text{exp_bin} = e\}_g \times Dist_g + X'_g \theta + \mu_b + \varepsilon_{gb},
\end{aligned} \tag{8}$$

where SC_Cand_{gb} is the number of SC candidates in GP g , $SCRes_g$ denotes SC reservation in 2016, $Dist_g$ is the running variable, X_g are GP-level controls, and μ_b are block fixed effects. We restrict the sample and include fixed effects to ensure like-with-like comparisons. First, block fixed effects absorb the administrative targeting of SHG expansion at the block level; conditional on these, the timing of SHG formation across GPs within a block was plausibly idiosyncratic. Second, we limit the analysis to GPs within the RD bandwidth to ensure comparable caste compositions. Table A28 shows that GPs with shorter and longer SHG exposure are similar across a wide range of pre-SHG characteristics, supporting the use of this residual variation as suggestive complementary evidence.²²

Figure A8 plots the corresponding coefficients. The core result replicates: SHG presence raises SC candidacy *only* where the GP is SC-reserved. The unreserved main effects (β_e) are near zero, whereas the reservation add-ons (δ_e) are positive. Second, there is *suggestive* evidence of a gradient in reserved GPs: the reservation add-on is larger for >3 years of exposure than for ≤ 3 years. Confidence intervals are wide and overlap, so we interpret this pattern cautiously. Consistent with the RCT, we find no gradient in unreserved GPs, suggesting that additional exposure does not translate into entry absent SC reservation. These results, though suggestive, provide valuable external validity to the RCT findings.

22. See also Table A29 for a comparison of never-exposed GPs with those that have some exposure.

Figure A8: RD at the SC-reservation cutoff: SC candidates (optimal bandwidth)



Notes: The figure shows local RD estimates of the number of Scheduled Caste (SC) candidates in the 2016 GP elections around the SC-reservation cutoff. The running variable is the GP's distance to the SC-reservation threshold in 2016; positive values indicate GPs reserved for SC. The optimal bandwidth is 500.95. Points and fitted lines are based on the specification in the text, with block fixed effects and robust standard errors. Units on the y -axis are candidates. Sample restricted to $|\text{distance}| < 500.95$.

XII. Construction of the Electability Index

For each candidate matched to the SECC, we observe three characteristics: years of education, drawn from candidate affidavits; the SECC asset score; and the size of the candidate’s sub-caste, proxied by the log prevalence of the candidate’s surname in the SECC. We standardize each characteristic relative to the distribution among candidates in control GPs. Section D examines GP-level averages of these characteristics, and Appendix IX describes the matching procedure.

To construct the index, we weight each characteristic by its ability to predict electoral victory, rather than imposing equal weights or assigning signs a priori. Col 1 of Table A30 reports a regression, estimated among the 502 candidates in control GPs, of an indicator for winning on the three standardized characteristics. The specification includes GP fixed effects, so the coefficients reflect comparisons among candidates contesting the same GP-head election. Because the weights are estimated using candidates in control GPs only, they cannot be affected by treatment.

The estimates in Col 1 are individually imprecise, given the size of the experimental sample. Cols 2 and 3 therefore examine whether the same characteristics predict victory in substantially larger samples constructed by matching the universe of candidates in the 2016 mukhiya elections to the SECC. Col 2 includes approximately 12,000 candidates in untreated GPs within the experiment’s seven districts, while Col 3 includes approximately 53,000 candidates across Bihar. The estimated weights on education and assets closely replicate those obtained from the experimental control group and are precisely estimated: a one-standard-deviation increase in either characteristic is associated with a 1.5–2.1 percentage point increase in the probability of victory, and the characteristics are jointly significant at $p < 0.001$. The estimated weight on sub-caste size is indistinguishable from zero in all three samples. We retain the weights estimated from the experimental control group for the main index; the larger samples provide evidence that the predictive relationships underlying these weights are not specific to the experimental sample.

The electability index is the weighted sum of the three standardized characteristics, re-standardized relative to the distribution among candidates in control GPs. It provides an empirical counterpart to valence in the framework developed in Section IV: a summary of the observed candidate characteristics that predict electoral success. Table A31 shows that the results in Section D are robust to alternative constructions of the index. When we instead use weights estimated from the district or statewide samples, or assign equal weights to the three components, the estimated decline in the average electability of the candidate pool in treated reserved GPs remains similar in magnitude. The group-scaled and instrumental-variables estimates are also statistically significant under each definition.

Table A30: Predicting Victory: Electability Weights Across Samples

	(1) Control GPs	(2) RCT districts	(3) Statewide
Education (SD)	0.0169 (0.0155)	0.0147*** (0.0034)	0.0169*** (0.0016)
Asset score (SD)	0.0139 (0.0188)	0.0157*** (0.0035)	0.0208*** (0.0017)
Sub-caste size, log (SD)	-0.0118 (0.0162)	-0.0000 (0.0032)	0.0019 (0.0017)
Joint significance (p)	0.541	<0.001	<0.001
Observations (candidates)	502	11947	53179
GPs	72	1570	7207
Mean of outcome	0.092	0.078	0.082

Notes: This table reports linear probability models in which the outcome is an indicator for winning the 2016 GP-head election. Regressors are standardized within each column's estimation sample. Col 1 uses candidates in the experiment's control GPs and includes the three characteristics used to construct the electability index: education, the SECC asset score, and log sub-caste size, proxied by surname prevalence in the SECC. Cols 2 and 3 use candidates from the universe of 2016 mukhiya elections who were matched to the SECC with a match score above 0.95. Col 2 includes candidates in untreated GPs within the experiment's seven districts, and Col 3 includes candidates throughout Bihar. In these columns, education is drawn from SEC affidavits, the asset score from the SECC, and sub-caste size is measured as the number of households belonging to the candidate's jati in the GP. All regressions include GP fixed effects, and standard errors are clustered at the GP level. The coefficients in Col 1 are used as the weights in the main electability index. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A31: Candidate-Pool Electability: Robustness to the Index Definition

	(1) Main: control-GP weights	(2) RCT-district weights	(3) Statewide weights	(4) Equal weights
Treated	0.140 (0.143)	0.099 (0.143)	0.090 (0.142)	0.043 (0.140)
Treated X SC Reserved	-0.512* (0.302)	-0.552 (0.353)	-0.549 (0.360)	-0.474 (0.374)
Groups/100 X SC Reserved	-0.450* (0.259)	-0.619** (0.308)	-0.621* (0.315)	-0.693** (0.322)
Groups/100 X SC Reserved (IV)	-1.052* (0.583)	-1.107* (0.620)	-1.096* (0.628)	-0.929 (0.632)
Observations	175	175	175	175

Notes: Each column reconstructs the electability index using the same three standardized components but a different set of weights. Column (1) uses the paper's main construction, based on weights estimated among control-GP candidates (Col 1 of Table A30). Columns (2) and (3) use weights estimated from the district and statewide samples, respectively, in Table A30. Col 4 assigns equal weights to the three components. Each column re-estimates the specifications reported in Section D: the treatment effect; the interaction between treatment and the GP's 2016 SC-reservation status; the interaction between the number of groups formed by 2015, measured in hundreds, and reservation status; and the corresponding instrumental-variables specification, in which the number of groups is instrumented by treatment. All specifications include GP controls and block fixed effects. Robust standard errors are reported; the instrumental-variables specifications use small-sample standard errors. The sample contains 175 GPs. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

XIII. Reservation rule Description

Bihar has reservation for SCs, STs, EBCs (*not* OBCs) and women. The rule for these is given below. The crucial takeaway from the sections below is that, focusing solely on SC reservation, there exists an SC population threshold below which no GP is reserved and above which a majority of GPs are reserved. Within this set of GPs, there exists another SC population threshold above which GPs are to be reserved for women. Note, also, that the rule for 2006 is fixed till 2016 (two election cycles), after which the reservation cycle switches. Below, we describe the reservation algorithm for caste/gender, first for 2006 and then for 2016.

A. 2006

The reservation rule proceeds in the following manner:

- First, based on the proportion of SCs (STs) in the block, the number of GPs to be reserved for SCs (STs) is decided. If there are N_j GPs in block j and θ_j is the proportion of SCs (STs) in block j , then the number of GPs, n_j , to be reserved is

$$n_j = \text{round}(\theta_j * N_j, 1)$$

- Let n_{SC} and n_{ST} be the number of GPs to be reserved in block j for SCs and STs, respectively. The number of GPs to be reserved for EBCs is given by

$$n_{EBC} = \min(\text{round}(0.2 * N_j, 1), \text{round}(0.5 * N_j - n_{SC} - n_{ST}, 1))$$

- If there are no STs in the block or n_{ST} is 0 (which is true in 480 of the 534 blocks), then the rule skips to the next step. However, if $n_{ST} > 0$, the rule proceeds by arranging all GPs in descending order of their ST population. The first GP in the list is then reserved for STs.
- Now, all remaining GPs are rearranged in the descending order of their non SCST population. The first GP on this truncated list is “blocked”. The choice of word is deliberate and conveys an important distinction: the GP is not “reserved”, it is merely blocked.
- Now, all unreserved and unblocked GPs are rearranged in descending order of their SC population. The first GP in this further truncated list is now reserved for SCs.
- This algorithm proceeds until the number of GPs reserved for STs = n_{ST} or the number reserved for SCs is n_{SC} . Once, a group hits its quota of reserved GPs, then the rearranging of GPs is no longer done by that group. For instance, if $n_{ST} = 1$, then, in the second round, GPs are no longer rearranged by ST population - instead, the rule proceeds straight to rearranging by non-SCST population.
- The algorithm further proceeds till the second group also hits its quota of reserved GPs. This throws up two sets of GPs, n_{ST} GPs that are reserved for STs and n_{SC} GPs that are reserved for SCs.

- Now, all the unreserved GPs (including the “blocked” ones) are collected and arranged by descending order of GP population.
- The first n_{EBC} GPs in this list is reserved for EBCs.
- Thus, for each block, one can arrive at an SC population cut-off - the SC population of the last GP to be reserved for SCs - below which no GP is reserved. This threshold varies by block. Figure A2 gives the first stage and shows that the first stage is robust to (a) adding block fixed effects (Panel (b)) and (b) dropping all ST/EBC reserved GPs.

A.1. Gender Reservation

- All the n_{SC} seats determined to be reserved for SCs within a block are arranged in descending order of SC population and *up to* 50% of them with the highest populations are reserved for SC women. For instance, if there are $n_{SC} = 3$, then 1 GP is reserved for SC women – the one with the highest SC population. On the other hand, if $n_{SC} = 4$, then 2 GPs are reserved for SC women – the top 2 highest SC population GPs.
- All the n_{ST} seats determined to be reserved for STs within a block are arranged in descending order of ST population and *up to* 50% of them with the highest ST populations are reserved for ST women.
- All the n_{EBC} seats determined to be reserved for EBCs within a block are arranged in descending order of total population and *up to* 50% of them with the highest total populations are reserved for EBC women.
- All the n_{GEN} seats determined to be caste unreserved within a block are arranged in descending order of non-SCST population and *up to* 50% of them with the highest non-SCST populations are reserved for women.

B. 2016

The reservation rule for 2016 proceeds in a similar manner to that of 2006, with two major changes. First, it changes the order in which GPs are arranged. In 2006, GPs were arranged first by STs, then non-SCSTs, then SCs. In 2016, GPs are arranged first by non-SCSTs, then by SCs, then by STs. Second, since there is no provision for recurring reservation, no GP previously reserved for SCs (STs/EBCs) can again be reserved for the same. So, when GPs are arranged by descending order of population of a particular group, those previously reserved are struck off the list, even before the algorithm begins.

C. A Worked Example

The following example, adapted from Kumar and Sharan (2025), traces the algorithm through a fictitious block of five GPs.

GP	SC pop.	ST pop.	Total pop.	Non-SC/ST pop.	Blocked	SC res.	Women res.	Running variable
1	1,000	0	10,000	9,000	✓		✓	−2,500
2	2,000	0	9,000	7,000	✓			−1,500
3	3,000	0	8,000	5,000				−500
4	4,000	0	7,000	3,000		✓		+500
5	5,000	250	6,000	1,000		✓	✓	+1,500

The block’s total population is 40,000, of which 15,000 is SC, so the number of SC-reserved GPs is $\text{round}((15,000/40,000) \times 5) = 2$; the ST population of 250 gives $n_{ST} = 0$. EBC reservation cannot exceed 20% of GPs, but because total caste reservation cannot exceed 50% of the block, no GP is reserved for EBCs here. The algorithm then proceeds:

- GPs are arranged in descending order of non-SC/ST population; the first, GP 1, is blocked.
- The remaining GPs are rearranged by SC population; the first, GP 5, is reserved for SCs.
- The remaining unblocked, unreserved GPs are again rearranged by non-SC/ST population; GP 2 is blocked.
- Rearranging the remainder by SC population reserves GP 4, exhausting the SC quota.
- Gender reservation follows. Among the two SC seats, *up to* 50% – one seat – goes to SC women: the SC-reserved GP with the higher SC population, GP 5. Among the three caste-unreserved GPs, *up to* 50% – one seat – is reserved for women: the GP with the highest non-SC/ST population, GP 1.

The SC populations of the last GP reserved for SCs (GP 4, at 4,000) and of the GP that narrowly missed (GP 3, at 3,000) bracket the block’s cut-off, set at the midpoint (3,500). The running variable in our RD designs is the GP’s SC population minus this block-specific cut-off: GP 4 sits at +500 and GP 3 at −500. Note also that the GPs on either side of the cut-off are both gender-unreserved: reservation for women does not jump at the SC threshold.